

T H I R D E D I T I O N

PRESSURE VESSEL DESIGN MANUAL

**Illustrated
procedures
for solving
major pressure
vessel design
problems**

DENNIS MOSS



THIRD EDITION
PRESSURE VESSEL
DESIGN MANUAL

THIRD EDITION

PRESSURE VESSEL DESIGN MANUAL

**Illustrated
procedures
for solving
major pressure
vessel design
problems**

DENNIS R. MOSS



ELSEVIER

**AMSTERDAM • BOSTON • HEIDELBERG
LONDON • NEW YORK • OXFORD • PARIS
SAN DIEGO • SAN FRANCISCO • SINGAPORE
SYDNEY • TOKYO**

**G | P Gulf
P | Professional
Publishing**

Gulf Professional Publishing is an imprint of Elsevier

Gulf Professional Publishing is an imprint of Elsevier
200 Wheeler Road, Burlington, MA 01803, USA
Linacre House, Jordan Hill, Oxford OX2 8DP, UK

Copyright © 2004, Elsevier, Inc. All rights reserved

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without the prior written permission of the publisher.

Permissions may be sought directly from Elsevier's Science & Technology Rights Department in Oxford, UK: phone (+44) 1865 843830, fax: (+44) 1865 853333, e-mail: permissions@elsevier.com.uk. You may also complete your request online via the Elsevier Science homepage (<http://elsevier.com>), by selecting "Customer Support" and then "Obtaining Permissions."



Recognizing the importance of preserving what has been written, Elsevier prints its books on acid-free paper whenever possible.

Library of Congress Cataloging-in-Publication Data

Moss, Dennis R.

Pressure vessel design manual: illustrated procedures for solving major pressure vessel design problems/Dennis R. Moss.—3rd ed.

p. cm.

ISBN 0-7506-7740-6 (hardcover: alk. paper)

1. Pressure vessels—Design and construction—Handbooks, manuals, etc. I. Title.

TA660.T34M68 2003

681'.76041—dc22

2003022552

British Library Cataloguing-in-Publication Data

A catalogue record for this book is available from the British Library.

ISBN: 0-7506-7740-6

For information on all Gulf Professional Publishing publications visit our website at www.gulfpp.com

04 05 06 07 08 10 11 9 8 7 6 5 4 3 2 1

Printed in the United States of America

Contents

PREFACE, ix

CHAPTER 1

STRESSES IN PRESSURE VESSELS, 1

Design Philosophy, 1
Stress Analysis, 1
Stress/Failure Theories, 2
Failures in Pressure Vessels, 5
Loadings, 6
Stress, 7
Special Problems, 10
References, 14

CHAPTER 2

GENERAL DESIGN, 15

Procedure 2-1: General Vessel Formulas, 15
Procedure 2-2: External Pressure Design, 19
Procedure 2-3: Calculate MAP, MAWP, and Test Pressures, 28
Procedure 2-4: Stresses in Heads Due to Internal Pressure, 30
Procedure 2-5: Design of Intermediate Heads, 31
Procedure 2-6: Design of Toriconical Transitions, 33
Procedure 2-7: Design of Flanges, 37
Procedure 2-8: Design of Spherically Dished Covers, 57
Procedure 2-9: Design of Blind Flanges with Openings, 58
Procedure 2-10: Bolt Torque Required for Sealing Flanges, 59
Procedure 2-11: Design of Flat Heads, 62
Procedure 2-12: Reinforcement for Studding Outlets, 68
Procedure 2-13: Design of Internal Support Beds, 69
Procedure 2-14: Nozzle Reinforcement, 74
Procedure 2-15: Design of Large Openings in Flat Heads, 78
Procedure 2-16: Find or Revise the Center of Gravity of a Vessel, 80
Procedure 2-17: Minimum Design Metal Temperature (MDMT), 81
Procedure 2-18: Buckling of Thin-Walled Cylindrical Shells, 85
Procedure 2-19: Optimum Vessel Proportions, 89
Procedure 2-20: Estimating Weights of Vessels and Vessel Components, 95
References, 106

CHAPTER 3

DESIGN OF VESSEL SUPPORTS, 109

Support Structures, 109

Procedure 3-1: Wind Design per ASCE, 112

Procedure 3-2: Wind Design per UBC-97, 118

Procedure 3-3: Seismic Design for Vessels, 120

Procedure 3-4: Seismic Design—Vessel on Unbraced Legs, 125

Procedure 3-5: Seismic Design—Vessel on Braced Legs, 132

Procedure 3-6: Seismic Design—Vessel on Rings, 140

Procedure 3-7: Seismic Design—Vessel on Lugs #1, 145

Procedure 3-8: Seismic Design—Vessel on Lugs #2, 151

Procedure 3-9: Seismic Design—Vessel on Skirt, 157

Procedure 3-10: Design of Horizontal Vessel on Saddles, 166

Procedure 3-11: Design of Saddle Supports for Large Vessels, 177

Procedure 3-12: Design of Base Plates for Legs, 184

Procedure 3-13: Design of Lug Supports, 188

Procedure 3-14: Design of Base Details for Vertical Vessels #1, 192

Procedure 3-15: Design of Base Details for Vertical Vessels #2, 200

References, 202

CHAPTER 4

SPECIAL DESIGNS, 203

Procedure 4-1: Design of Large-Diameter Nozzle Openings, 203

Procedure 4-2: Design of Cone–Cylinder Intersections, 208

Procedure 4-3: Stresses at Circumferential Ring Stiffeners, 216

Procedure 4-4: Tower Deflection, 219

Procedure 4-5: Design of Ring Girders, 222

Procedure 4-6: Design of Baffles, 227

Procedure 4-7: Design of Vessels with Refractory Linings, 237

Procedure 4-8: Vibration of Tall Towers and Stacks, 244

References, 254

CHAPTER 5

LOCAL LOADS, 255

Procedure 5-1: Stresses in Circular Rings, 256

Procedure 5-2: Design of Partial Ring Stiffeners, 265

Procedure 5-3: Attachment Parameters, 267

Procedure 5-4: Stresses in Cylindrical Shells from External Local Loads, 269

Procedure 5-5: Stresses in Spherical Shells from External Local Loads, 283

References, 290

CHAPTER 6

RELATED EQUIPMENT, 291

Procedure 6-1: Design of Davits, 291

Procedure 6-2: Design of Circular Platforms, 296

Procedure 6-3: Design of Square and Rectangular Platforms, 304
 Procedure 6-4: Design of Pipe Supports, 309
 Procedure 6-5: Shear Loads in Bolted Connections, 317
 Procedure 6-6: Design of Bins and Elevated Tanks, 318
 Procedure 6-7: Agitators/Mixers for Vessels and Tanks, 328
 Procedure 6-8: Design of Pipe Coils for Heat Transfer, 335
 Procedure 6-9: Field-Fabricated Spheres, 355
 References, 364

CHAPTER 7

TRANSPORTATION AND ERECTION OF PRESSURE VESSELS, 365

Procedure 7-1: Transportation of Pressure Vessels, 365
 Procedure 7-2: Erection of Pressure Vessels, 387
 Procedure 7-3: Lifting Attachments and Terminology, 391
 Procedure 7-4: Lifting Loads and Forces, 400
 Procedure 7-5: Design of Tail Beams, Lugs, and Base Ring Details, 406
 Procedure 7-6: Design of Top Head and Cone Lifting Lugs, 416
 Procedure 7-7: Design of Flange Lugs, 420
 Procedure 7-8: Design of Trunnions, 431
 Procedure 7-9: Local Loads in Shell Due to Erection Forces, 434
 Procedure 7-10: Miscellaneous, 437

APPENDICES, 443

Appendix A: Guide to ASME Section VIII, Division 1, 443
 Appendix B: Design Data Sheet for Vessels, 444
 Appendix C: Joint Efficiencies (ASME Code), 445
 Appendix D: Properties of Heads, 447
 Appendix E: Volumes and Surface Areas of Vessel Sections, 448
 Appendix F: Vessel Nomenclature, 455
 Appendix G: Useful Formulas for Vessels, 459
 Appendix H: Material Selection Guide, 464
 Appendix I: Summary of Requirements for 100% X-Ray and PWHT, 465
 Appendix J: Material Properties, 466
 Appendix K: Metric Conversions, 474
 Appendix L: Allowable Compressive Stress for Columns, F_A , 475
 Appendix M: Design of Flat Plates, 478
 Appendix N: External Insulation for Vertical Vessels, 480
 Appendix O: Flow over Weirs, 482
 Appendix P: Time Required to Drain Vessels, 483
 Appendix Q: Vessel Surge Capacities and Hold-Up Times, 485
 Appendix R: Minor Defect Evaluation Procedure, 486
 References, 487
 Index, 489

Preface

Designers of pressure vessels and related equipment frequently have design information scattered among numerous books, periodicals, journals, and old notes. Then, when faced with a particular problem, they spend hours researching its solution only to discover the execution may have been rather simple. This book can eliminate those hours of research by providing a step-by-step approach to the problems most frequently encountered in the design of pressure vessels.

This book makes no claim to originality other than that of format. The material is organized in the most concise and functionally useful manner. Whenever possible, credit has been given to the original sources.

Although every effort has been made to obtain the most accurate data and solutions, it is the nature of engineering that certain simplifying assumptions be made. Solutions achieved should be viewed in this light, and where judgments are required, they should be made with due consideration.

Many experienced designers will have already performed many of the calculations outlined in this book, but will find the approach slightly different. All procedures have been developed and proven, using actual design problems. The procedures are easily repeatable to ensure consistency of execution. They also can be modified to incorporate changes in codes, standards, contracts, or local requirements. Everything required for the solution of an individual problem is contained in the procedure.

This book may be used directly to solve problems, as a guideline, as a logical approach to problems, or as a check to alternative design methods. If more detailed solutions are required, the approach shown can be amplified where required.

The user of this book should be advised that any code formulas or references should always be checked against the latest editions of codes, i.e., ASME Section VIII, Division 1, Uniform Building Code, and ASCE 7-95. These codes are continually updated and revised to incorporate the latest available data.

I am grateful to all those who have contributed information and advice to make this book possible, and invite any suggestions readers may make concerning corrections or additions.

Dennis R. Moss

Cover Photo: Photo courtesy of Irving Oil Ltd., Saint John, New Brunswick, Canada and Stone and Webster, Inc., A Shaw Group Company, Houston, Texas. The photo shows the Reactor-Regenerator Structure of the Converter Section of the RFCC (Resid Fluid Catalytic Cracking) Unit. This “world class” unit operates at the Irving Refinery Complex in Saint John, New Brunswick, Canada, and is a proprietary process of Stone and Webster.

1 Stresses in Pressure Vessels

DESIGN PHILOSOPHY

In general, pressure vessels designed in accordance with the ASME Code, Section VIII, Division 1, are designed by rules and do not require a detailed evaluation of all stresses. It is recognized that high localized and secondary bending stresses may exist but are allowed for by use of a higher safety factor and design rules for details. It is required, however, that all loadings (the forces applied to a vessel or its structural attachments) must be considered. (See Reference 1, Para. UG-22.)

While the Code gives formulas for thickness and stress of basic components, it is up to the designer to select appropriate analytical procedures for determining stress due to other loadings. The designer must also select the most probable combination of simultaneous loads for an economical and safe design.

The Code establishes allowable stresses by stating in Para. UG-23(c) that the maximum general primary membrane stress must be less than allowable stresses outlined in material sections. Further, it states that the maximum primary membrane stress *plus* primary bending stress may not exceed 1.5 times the allowable stress of the material sections. In other sections, specifically Paras. 1-5(e) and 2-8, higher allowable stresses are permitted if appropriate analysis is made. These higher allowable stresses clearly indicate that different stress levels for different stress categories are acceptable.

It is general practice when doing more detailed stress analysis to apply higher allowable stresses. In effect, the detailed evaluation of stresses permits substituting knowledge of localized stresses and the use of higher allowables in place of the larger factor of safety used by the Code. This higher safety factor really reflected lack of knowledge about actual stresses.

A calculated value of stress means little until it is associated with its location and distribution in the vessel and with the type of loading that produced it. Different types of stress have different degrees of significance.

The designer must familiarize himself with the various types of stress and loadings in order to accurately apply the results of analysis. The designer must also consider some adequate stress or failure theory in order to combine stresses and set allowable stress limits. It is against this failure mode that he must compare and interpret stress values, and define how the stresses in a component react and contribute to the strength of that part.

The following sections will provide the fundamental knowledge for applying the results of analysis. The topics covered in Chapter 1 form the basis by which the rest of the book is to be used. A section on special problems and considerations is included to alert the designer to more complex problems that exist.

STRESS ANALYSIS

Stress analysis is the determination of the relationship between external forces applied to a vessel and the corresponding stress. The emphasis of this book is not how to do stress analysis in particular, but rather how to analyze vessels and their component parts in an effort to arrive at an economical and safe design—the difference being that we analyze stresses where necessary to determine thickness of material and sizes of members. We are not so concerned with building mathematical models as with providing a step-by-step approach to the design of ASME Code vessels. It is not necessary to find every stress but rather to know the

governing stresses and how they relate to the vessel or its respective parts, attachments, and supports.

The starting place for stress analysis is to determine all the design conditions for a given problem and then determine all the related external forces. We must then relate these external forces to the vessel parts which must resist them to find the corresponding stresses. By isolating the causes (loadings), the effects (stress) can be more accurately determined.

The designer must also be keenly aware of the types of loads and how they relate to the vessel as a whole. Are the

effects long or short term? Do they apply to a localized portion of the vessel or are they uniform throughout?

How these stresses are interpreted and combined, what significance they have to the overall safety of the vessel, and what allowable stresses are applied will be determined by three things:

1. The strength/failure theory utilized.
2. The types and categories of loadings.
3. The hazard the stress represents to the vessel.

Membrane Stress Analysis

Pressure vessels commonly have the form of spheres, cylinders, cones, ellipsoids, tori, or composites of these. When the thickness is small in comparison with other dimensions ($R_m/t > 10$), vessels are referred to as membranes and the associated stresses resulting from the contained pressure are called membrane stresses. These membrane stresses are average tension or compression stresses. They are assumed to be uniform across the vessel wall and act tangentially to its surface. The membrane or wall is assumed to offer no resistance to bending. When the wall offers resistance to bending, bending stresses occur in addition to membrane stresses.

In a vessel of complicated shape subjected to internal pressure, the simple membrane-stress concepts do not suffice to give an adequate idea of the true stress situation. The types of heads closing the vessel, effects of supports, variations in thickness and cross section, nozzles, external attachments, and overall bending due to weight, wind, and seismic activity all cause varying stress distributions in the vessel. Deviations from a true membrane shape set up bending in the vessel wall and cause the direct loading to vary from point to point. The direct loading is diverted from the more flexible to the more rigid portions of the vessel. This effect is called "stress redistribution."

In any pressure vessel subjected to internal or external pressure, stresses are set up in the shell wall. The state of stress is triaxial and the three principal stresses are:

$$\begin{aligned}\sigma_x &= \text{longitudinal/meridional stress} \\ \sigma_\phi &= \text{circumferential/latitudinal stress} \\ \sigma_r &= \text{radial stress}\end{aligned}$$

In addition, there may be bending and shear stresses. The radial stress is a direct stress, which is a result of the pressure acting directly on the wall, and causes a compressive stress equal to the pressure. In thin-walled vessels this stress is so small compared to the other "principal" stresses that it is generally ignored. Thus we assume for purposes of analysis that the state of stress is biaxial. This greatly simplifies the method of combining stresses in comparison to triaxial stress states. For thickwalled vessels ($R_m/t < 10$), the radial stress cannot be ignored and formulas are quite different from those used in finding "membrane stresses" in thin shells.

Since ASME Code, Section VIII, Division 1, is basically for design by rules, a higher factor of safety is used to allow for the "unknown" stresses in the vessel. This higher safety factor, which allows for these unknown stresses, can impose a penalty on design but requires much less analysis. The design techniques outlined in this text are a compromise between finding all stresses and utilizing minimum code formulas. This additional knowledge of stresses warrants the use of higher allowable stresses in some cases, while meeting the requirements that all loadings be considered.

In conclusion, "membrane stress analysis" is not completely accurate but allows certain simplifying assumptions to be made while maintaining a fair degree of accuracy. The main simplifying assumptions are that the stress is biaxial and that the stresses are uniform across the shell wall. For thin-walled vessels these assumptions have proven themselves to be reliable. No vessel meets the criteria of being a true membrane, but we can use this tool with a reasonable degree of accuracy.

STRESS/FAILURE THEORIES

As stated previously, stresses are meaningless until compared to some stress/failure theory. The significance of a given stress must be related to its location in the vessel and its bearing on the ultimate failure of that vessel. Historically, various "theories" have been derived to combine and measure stresses against the potential failure mode. A number of stress theories, also called "yield criteria," are available for describing the effects of combined stresses. For purposes of this book, as these failure theories apply to pressure vessels, only two theories will be discussed.

They are the "maximum stress theory" and the "maximum shear stress theory."

Maximum Stress Theory

This theory is the oldest, most widely used and simplest to apply. Both ASME Code, Section VIII, Division 1, and Section I use the maximum stress theory as a basis for design. This theory simply asserts that the breakdown of

material depends only on the numerical magnitude of the maximum principal or normal stress. Stresses in the other directions are disregarded. Only the maximum principal stress must be determined to apply this criterion. This theory is used for biaxial states of stress assumed in a thin-walled pressure vessel. As will be shown later it is unconservative in some instances and requires a higher safety factor for its use. While the maximum stress theory does accurately predict failure in brittle materials, it is not always accurate for ductile materials. Ductile materials often fail along lines 45° to the applied force by shearing, long before the tensile or compressive stresses are maximum.

This theory can be illustrated graphically for the four states of biaxial stress shown in Figure 1-1.

It can be seen that uniaxial tension or compression lies on the two axes. Inside the box (outer boundaries) is the elastic range of the material. Yielding is predicted for stress combinations by the outer line.

Maximum Shear Stress Theory

This theory asserts that the breakdown of material depends only on the maximum shear stress attained in an element. It assumes that yielding starts in planes of maximum shear stress. According to this theory, yielding will start at a point when the maximum shear stress at that point reaches one-half of the uniaxial yield strength, F_y . Thus for a

biaxial state of stress where $\sigma_1 > \sigma_2$, the maximum shear stress will be $(\sigma_1 - \sigma_2)/2$.

Yielding will occur when

$$\frac{\sigma_1 - \sigma_2}{2} = \frac{F_y}{2}$$

Both ASME Code, Section VIII, Division 2 and ASME Code, Section III, utilize the maximum shear stress criterion. This theory closely approximates experimental results and is also easy to use. This theory also applies to triaxial states of stress. In a triaxial stress state, this theory predicts that yielding will occur whenever one-half the algebraic difference between the maximum and minimum stress is equal to one-half the yield stress. Where $\sigma_1 > \sigma_2 > \sigma_3$, the maximum shear stress is $(\sigma_1 - \sigma_3)/2$.

Yielding will begin when

$$\frac{\sigma_1 - \sigma_3}{2} = \frac{F_y}{2}$$

This theory is illustrated graphically for the four states of biaxial stress in Figure 1-2.

A comparison of Figure 1-1 and Figure 1-2 will quickly illustrate the major differences between the two theories. Figure 1-2 predicts yielding at earlier points in Quadrants II and IV. For example, consider point B of Figure 1-2. It shows $\sigma_2 = (-)\sigma_1$; therefore the shear stress is equal to $\sigma_2 - (-\sigma_1)/2$, which equals $\sigma_2 + \sigma_1/2$ or one-half the stress

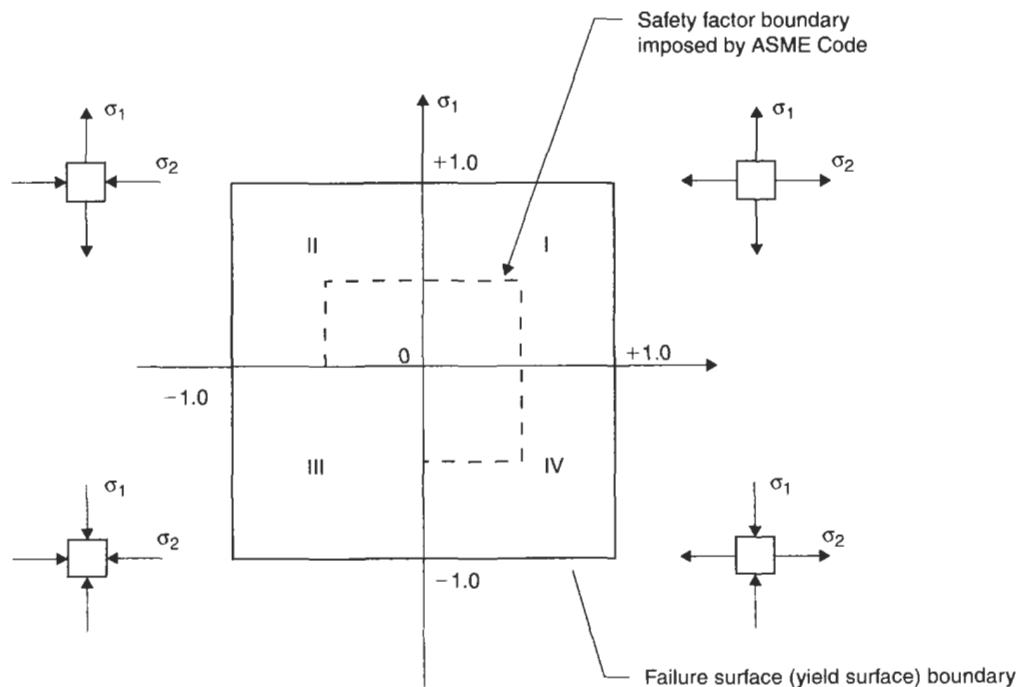


Figure 1-1. Graph of maximum stress theory. Quadrant I: biaxial tension; Quadrant II: tension; Quadrant III: biaxial compression; Quadrant IV: compression.

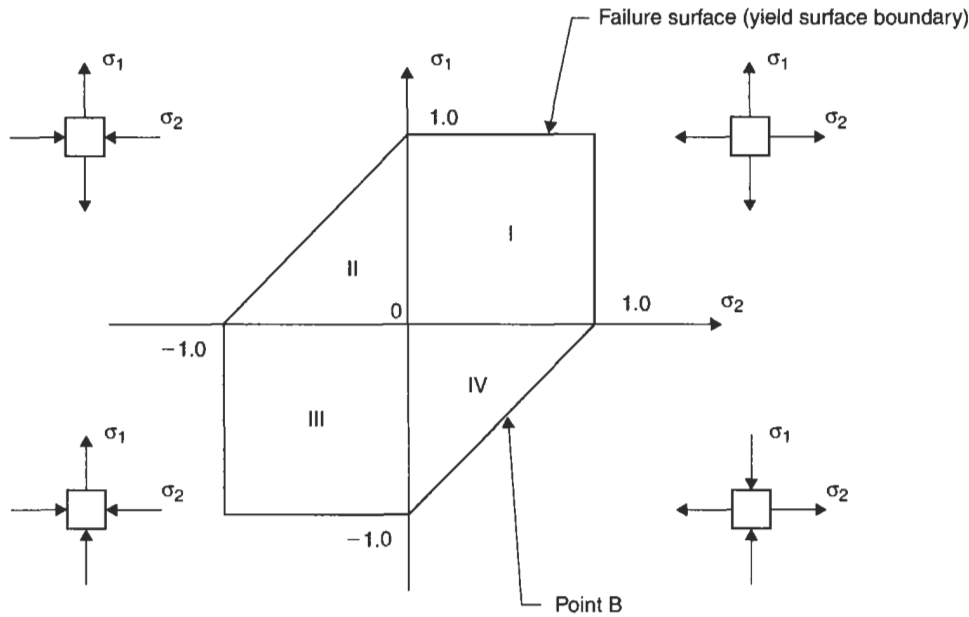


Figure 1-2. Graph of maximum shear stress theory.

which would cause yielding as predicted by the maximum stress theory!

Comparison of the Two Theories

Both theories are in agreement for uniaxial stress or when one of the principal stresses is large in comparison to the others. The discrepancy between the theories is greatest when both principal stresses are numerically equal.

For simple analysis upon which the thickness formulas for ASME Code, Section I or Section VIII, Division 1, are based, it makes little difference whether the maximum stress theory or maximum shear stress theory is used. For example, according to the maximum stress theory, the controlling stress governing the thickness of a cylinder is σ_ϕ , circumferential stress, since it is the largest of the three principal stresses. According to the maximum shear stress theory, the controlling stress would be one-half the algebraic difference between the maximum and minimum stress:

- The maximum stress is the circumferential stress, σ_ϕ

$$\sigma_\phi = PR/t$$

- The minimum stress is the radial stress, σ_r

$$\sigma_r = -P$$

Therefore, the maximum shear stress is:

$$\frac{\sigma_\phi - \sigma_r}{2}$$

ASME Code, Section VIII, Division 2, and Section III use the term “stress intensity,” which is defined as twice the maximum shear stress. Since the shear stress is compared to one-half the yield stress only, “stress intensity” is used for comparison to allowable stresses or ultimate stresses. To define it another way, yielding begins when the “stress intensity” exceeds the yield strength of the material.

In the preceding example, the “stress intensity” would be equal to $\sigma_\phi - \sigma_r$. And

$$\sigma_\phi - \sigma_r = PR/t - (-P) = PR/t + P$$

For a cylinder where $P = 300$ psi, $R = 30$ in., and $t = .5$ in., the two theories would compare as follows:

- *Maximum stress theory*

$$\sigma = \sigma_\phi = PR/t = 300(30)/.5 = 18,000 \text{ psi}$$

- *Maximum shear stress theory*

$$\sigma = PR/t + P = 300(30)/.5 + 300 = 18,300 \text{ psi}$$

Two points are obvious from the foregoing:

1. For thin-walled pressure vessels, both theories yield approximately the same results.
2. For thin-walled pressure vessels the radial stress is so small in comparison to the other principal stresses that it can be ignored and a state of biaxial stress is assumed to exist.

For thick-walled vessels ($R_n/t < 10$), the radial stress becomes significant in defining the ultimate failure of the vessel. The maximum stress theory is unconservative for

designing these vessels. For this reason, this text has limited its application to thin-walled vessels where a biaxial state of stress is assumed to exist.

FAILURES IN PRESSURE VESSELS

Vessel failures can be grouped into four major categories, which describe *why* a vessel failure occurs. Failures can also be grouped into types of failures, which describe *how* the failure occurs. Each failure has a why and how to its history. It may have failed *through* corrosion fatigue *because* the wrong material was selected! The designer must be as familiar with categories and types of failure as with categories and types of stress and loadings. Ultimately they are all related.

Categories of Failures

1. *Material*—Improper selection of material; defects in material.
2. *Design*—Incorrect design data; inaccurate or incorrect design methods; inadequate shop testing.
3. *Fabrication*—Poor quality control; improper or insufficient fabrication procedures including welding; heat treatment or forming methods.
4. *Service*—Change of service condition by the user; inexperienced operations or maintenance personnel; upset conditions. Some types of service which require special attention both for selection of material, design details, and fabrication methods are as follows:
 - a. Lethal
 - b. Fatigue (cyclic)
 - c. Brittle (low temperature)
 - d. High temperature
 - e. High shock or vibration
 - f. Vessel contents
 - Hydrogen
 - Ammonia
 - Compressed air
 - Caustic
 - Chlorides

Types of Failures

1. *Elastic deformation*—Elastic instability or elastic buckling, vessel geometry, and stiffness as well as properties of materials are protection against buckling.

2. *Brittle fracture*—Can occur at low or intermediate temperatures. Brittle fractures have occurred in vessels made of low carbon steel in the 40°–50°F range during hydrotest where minor flaws exist.
3. *Excessive plastic deformation*—The primary and secondary stress limits as outlined in ASME Section VIII, Division 2, are intended to prevent excessive plastic deformation and incremental collapse.
4. *Stress rupture*—Creep deformation as a result of fatigue or cyclic loading, i.e., progressive fracture. Creep is a time-dependent phenomenon, whereas fatigue is a cycle-dependent phenomenon.
5. *Plastic instability*—Incremental collapse; incremental collapse is cyclic strain accumulation or cumulative cyclic deformation. Cumulative damage leads to instability of vessel by plastic deformation.
6. *High strain*—Low cycle fatigue is strain-governed and occurs mainly in lower-strength/high-ductile materials.
7. *Stress corrosion*—It is well known that chlorides cause stress corrosion cracking in stainless steels; likewise caustic service can cause stress corrosion cracking in carbon steels. Material selection is critical in these services.
8. *Corrosion fatigue*—Occurs when corrosive and fatigue effects occur simultaneously. Corrosion can reduce fatigue life by pitting the surface and propagating cracks. Material selection and fatigue properties are the major considerations.

In dealing with these various modes of failure, the designer must have at his disposal a picture of the state of stress in the various parts. It is against these failure modes that the designer must compare and interpret stress values. But setting allowable stresses is not enough! For elastic instability one must consider geometry, stiffness, and the properties of the material. Material selection is a major consideration when related to the type of service. Design details and fabrication methods are as important as “allowable stress” in design of vessels for cyclic service. The designer and all those persons who ultimately affect the design must have a clear picture of the conditions under which the vessel will operate.

LOADINGS

Loadings or forces are the “causes” of stresses in pressure vessels. These forces and moments must be isolated both to determine *where* they apply to the vessel and *when* they apply to a vessel. Categories of loadings define where these forces are applied. Loadings may be applied over a large portion (general area) of the vessel or over a local area of the vessel. Remember both *general* and *local* loads can produce membrane and bending stresses. These stresses are additive and define the overall state of stress in the vessel or component. Stresses from local loads must be added to stresses from general loadings. These combined stresses are then compared to an allowable stress.

Consider a pressurized, vertical vessel bending due to wind, which has an inward radial force applied locally. The effects of the pressure loading are longitudinal and circumferential tension. The effects of the wind loading are longitudinal tension on the windward side and longitudinal compression on the leeward side. The effects of the local inward radial load are some local membrane stresses and local bending stresses. The local stresses would be both circumferential and longitudinal, tension on the inside surface of the vessel, and compressive on the outside. Of course the steel at any given point only sees a certain level of stress or the combined effect. It is the designer's job to combine the stresses from the various loadings to arrive at the worst probable combination of stresses, combine them using some failure theory, and compare the results to an acceptable stress level to obtain an economical and safe design.

This hypothetical problem serves to illustrate how categories and types of loadings are related to the stresses they produce. The stresses applied more or less *continuously* and *uniformly* across an entire section of the vessel are primary stresses.

The stresses due to pressure and wind are primary membrane stresses. These stresses should be limited to the code allowable. These stresses would cause the bursting or collapse of the vessel if allowed to reach an unacceptably high level.

On the other hand, the stresses from the inward radial load could be either a primary local stress or secondary stress. It is a primary local stress if it is produced from an unrelenting load or a secondary stress if produced by a relenting load. Either stress may cause local deformation but will not in and of itself cause the vessel to fail. If it is a primary stress, the stress will be redistributed; if it is a secondary stress, the load will relax once slight deformation occurs.

Also be aware that this is only true for ductile materials. In brittle materials, there would be no difference between

primary and secondary stresses. If the material cannot yield to reduce the load, then the definition of secondary stress does not apply! Fortunately current pressure vessel codes require the use of ductile materials.

This should make it obvious that the type and category of loading will determine the type and category of stress. This will be expanded upon later, but basically each combination of stresses (stress categories) will have different allowables, i.e.:

- Primary stress: $P_m < SE$
- Primary membrane local (P_L):

$$P_L = P_m + P_L < 1.5 SE$$

$$P_L = P_m + Q_m < 1.5 SE$$

- Primary membrane + secondary (Q):

$$P_m + Q < 3 SE$$

But what if the loading was of relatively short duration? This describes the “type” of loading. Whether a loading is steady, more or less continuous, or nonsteady, variable, or temporary will also have an effect on what level of stress will be acceptable. If in our hypothetical problem the loading had been pressure + seismic + local load, we would have a different case. Due to the relatively short duration of seismic loading, a higher “temporary” allowable stress would be acceptable. The vessel doesn't have to operate in an earthquake all the time. On the other hand, it also shouldn't fall down in the event of an earthquake! Structural designs allow a one-third increase in allowable stress for seismic loadings for this reason.

For *steady loads*, the vessel must support these loads more or less continuously during its useful life. As a result, the stresses produced from these loads must be maintained to an acceptable level.

For *nonsteady loads*, the vessel may experience some or all of these loadings at various times but not all at once and not more or less continuously. Therefore a temporarily higher stress is acceptable.

For *general loads* that apply more or less uniformly across an entire section, the corresponding stresses must be lower, since the entire vessel must support that loading.

For *local loads*, the corresponding stresses are confined to a small portion of the vessel and normally fall off rapidly in distance from the applied load. As discussed previously, pressurizing a vessel causes bending in certain components. But it doesn't cause the entire vessel to bend. The results are not as significant (except in cyclic service) as those caused by general loadings. Therefore a slightly higher allowable stress would be in order.

Loadings can be outlined as follows:

A. Categories of loadings

1. *General loads*—Applied more or less continuously across a vessel section.
 - a. Pressure loads—Internal or external pressure (design, operating, hydrotest, and hydrostatic head of liquid).
 - b. Moment loads—Due to wind, seismic, erection, transportation.
 - c. Compressive/tensile loads—Due to dead weight, installed equipment, ladders, platforms, piping, and vessel contents.
 - d. Thermal loads—Hot box design of skirthead attachment.
2. *Local loads*—Due to reactions from supports, internals, attached piping, attached equipment, i.e., platforms, mixers, etc.
 - a. Radial load—Inward or outward.
 - b. Shear load—Longitudinal or circumferential.
 - c. Torsional load.

- d. Tangential load.
- e. Moment load—Longitudinal or circumferential.
- f. Thermal load.

B. Types of loadings

1. *Steady loads*—Long-term duration, continuous.
 - a. Internal/external pressure.
 - b. Dead weight.
 - c. Vessel contents.
 - d. Loadings due to attached piping and equipment.
 - e. Loadings to and from vessel supports.
 - f. Thermal loads.
 - g. Wind loads.
2. *Nonsteady loads*—Short-term duration; variable.
 - a. Shop and field hydrotests.
 - b. Earthquake.
 - c. Erection.
 - d. Transportation.
 - e. Upset, emergency.
 - f. Thermal loads.
 - g. Start up, shut down.

STRESS

ASME Code, Section VIII, Division 1 vs. Division 2

ASME Code, Section VIII, Division 1 does not explicitly consider the effects of combined stress. Neither does it give detailed methods on how stresses are combined. ASME Code, Section VIII, Division 2, on the other hand, provides specific guidelines for stresses, how they are combined, and allowable stresses for categories of combined stresses. Division 2 is design by analysis whereas Division 1 is design by rules. Although stress analysis as utilized by Division 2 is beyond the scope of this text, the use of stress categories, definitions of stress, and allowable stresses is applicable.

Division 2 stress analysis considers all stresses in a triaxial state combined in accordance with the maximum shear stress theory. Division 1 and the procedures outlined in this book consider a biaxial state of stress combined in accordance with the maximum stress theory. Just as you would not design a nuclear reactor to the rules of Division 1, you would not design an air receiver by the techniques of Division 2. Each has its place and applications. The following discussion on categories of stress and allowables will utilize information from Division 2, which can be applied in general to all vessels.

Types, Classes, and Categories of Stress

The shell thickness as computed by Code formulas for internal or external pressure alone is often not sufficient to withstand the combined effects of all other loadings. Detailed calculations consider the effects of each loading separately and then must be combined to give the total state of stress in that part. The stresses that are present in pressure vessels are separated into various *classes* in accordance with the *types* of loads that produced them, and the hazard they represent to the vessel. Each class of stress must be maintained at an acceptable level and the combined total stress must be kept at another acceptable level. The combined stresses due to a combination of loads acting simultaneously are called stress categories. Please note that this terminology differs from that given in Division 2, but is clearer for the purposes intended here.

Classes of stress, categories of stress, and allowable stresses are based on the type of loading that produced them *and* on the hazard they represent to the structure. Unrelenting loads produce primary stresses. Relenting loads (self-limiting) produce secondary stresses. General loadings produce primary membrane and bending stresses. Local loads produce local membrane and bending stresses. Primary stresses must be kept lower than secondary stresses.

Primary plus secondary stresses are allowed to be higher and so on. Before considering the combination of stresses (categories), we must first define the various *types* and *classes* of stress.

Types of Stress

There are many names to describe types of stress. Enough in fact to provide a confusing picture even to the experienced designer. As these stresses apply to pressure vessels, we group all types of stress into three major classes of stress, and subdivision of each of the groups is arranged according to their effect on the vessel. The following list of stresses describes types of stress without regard to their effect on the vessel or component. They define a direction of stress or relate to the application of the load.

- | | |
|------------------|---------------------|
| 1. Tensile | 10. Thermal |
| 2. Compressive | 11. Tangential |
| 3. Shear | 12. Load induced |
| 4. Bending | 13. Strain induced |
| 5. Bearing | 14. Circumferential |
| 6. Axial | 15. Longitudinal |
| 7. Discontinuity | 16. Radial |
| 8. Membrane | 17. Normal |
| 9. Principal | |

Classes of Stress

The foregoing list provides examples of types of stress. It is, however, too general to provide a basis with which to combine stresses or apply allowable stresses. For this purpose, new groupings called *classes* of stress must be used. Classes of stress are defined by the type of loading which produces them and the hazard they represent to the vessel.

1. *Primary stress*
 - a. General:
 - Primary general membrane stress, P_m
 - Primary general bending stress, P_b
 - b. Primary local stress, P_L
2. *Secondary stress*
 - a. Secondary membrane stress, Q_m
 - b. Secondary bending stress, Q_b
3. *Peak stress*, F

Definitions and examples of these stresses follow.

Primary general stress. These stresses act over a full cross section of the vessel. They are produced by mechanical loads (load induced) and are the most hazardous of all types of stress. The basic characteristic of a primary stress is that it

is not self-limiting. Primary stresses are generally due to internal or external pressure or produced by sustained external forces and moments. Thermal stresses are never classified as primary stresses.

Primary general stresses are divided into membrane and bending stresses. The need for dividing primary general stress into membrane and bending is that the calculated value of a primary bending stress may be allowed to go higher than that of a primary membrane stress. Primary stresses that exceed the yield strength of the material can cause failure or gross distortion. Typical calculations of primary stress are:

$$\frac{PR}{t}, \frac{F}{A}, \frac{MC}{I}, \text{ and } \frac{TC}{J}$$

Primary general membrane stress, P_m . This stress occurs across the entire cross section of the vessel. It is remote from discontinuities such as head-shell intersections, cone-cylinder intersections, nozzles, and supports. Examples are:

- a. Circumferential and longitudinal stress due to pressure.
- b. Compressive and tensile axial stresses due to wind.
- c. Longitudinal stress due to the bending of the horizontal vessel over the saddles.
- d. Membrane stress in the center of the flat head.
- e. Membrane stress in the nozzle wall within the area of reinforcement due to pressure or external loads.
- f. Axial compression due to weight.

Primary general bending stress, P_b . Primary bending stresses are due to sustained loads and are capable of causing collapse of the vessel. There are relatively few areas where primary bending occurs:

- a. Bending stress in the center of a flat head or crown of a dished head.
- b. Bending stress in a shallow conical head.
- c. Bending stress in the ligaments of closely spaced openings.

Local primary membrane stress, P_L . Local primary membrane stress is not technically a classification of stress but a stress category, since it is a combination of two stresses. The combination it represents is primary membrane stress, P_m , plus secondary membrane stress, Q_m , produced from sustained loads. These have been grouped together in order to limit the allowable stress for this particular combination to a level lower than allowed for other primary and secondary stress applications. It was felt that local stress from sustained (unrelenting) loads presented a great enough hazard for the combination to be "classified" as a primary stress.

A local primary stress is produced either by design pressure alone or by other mechanical loads. Local primary

stresses have some self-limiting characteristics like secondary stresses. Since they are localized, once the yield strength of the material is reached, the load is redistributed to stiffer portions of the vessel. However, since any deformation associated with yielding would be unacceptable, an allowable stress lower than secondary stresses is assigned. The basic difference between a primary local stress and a secondary stress is that a primary local stress is produced by a load that is unrelenting; the stress is just redistributed. In a secondary stress, yielding relaxes the load and is truly self-limiting. The ability of primary local stresses to redistribute themselves after the yield strength is attained locally provides a safety-valve effect. Thus, the higher allowable stress applies only to a local area.

Primary local membrane stresses are a combination of membrane stresses only. Thus only the "membrane" stresses from a local load are combined with primary general membrane stresses, not the bending stresses. The bending stresses associated with a local loading are secondary stresses. Therefore, the membrane stresses from a WRC-107-type analysis must be broken out separately and combined with primary general stresses. The same is true for discontinuity membrane stresses at head-shell junctures, cone-cylinder junctures, and nozzle-shell junctures. The bending stresses would be secondary stresses.

Therefore, $P_L = P_m + Q_m$, where Q_m is a local stress from a sustained or unrelenting load. Examples of primary local membrane stresses are:

- a. P_m + membrane stresses at local discontinuities:
 1. Head-shell juncture
 2. Cone-cylinder juncture
 3. Nozzle-shell juncture
 4. Shell-flange juncture
 5. Head-skirt juncture
 6. Shell-stiffening ring juncture
- b. P_m + membrane stresses from local sustained loads:
 1. Support lugs
 2. Nozzle loads
 3. Beam supports
 4. Major attachments

Secondary stress. The basic characteristic of a secondary stress is that it is self-limiting. As defined earlier, this means that local yielding and minor distortions can satisfy the conditions which caused the stress to occur. Application of a secondary stress cannot cause structural failure due to the restraints offered by the body to which the part is attached. Secondary mean stresses are developed at the junctions of major components of a pressure vessel. Secondary mean stresses are also produced by sustained loads other than internal or external pressure. Radial loads on nozzles produce secondary mean stresses in the shell at the junction of the nozzle. Secondary stresses are strain-induced stresses.

Discontinuity stresses are only considered as secondary stresses if their extent along the length of the shell is limited. Division 2 imposes the restriction that the length over which the stress is secondary is $\sqrt{R_m t}$. Beyond this distance, the stresses are considered as primary mean stresses. In a cylindrical vessel, the length $\sqrt{R_m t}$ represents the length over which the shell behaves as a ring.

A further restriction on secondary stresses is that they may not be closer to another gross structural discontinuity than a distance of $2.5\sqrt{R_m t}$. This restriction is to eliminate the additive effects of edge moments and forces.

Secondary stresses are divided into two additional groups, membrane and bending. Examples of each are as follows:

Secondary membrane stress, Q_m .

- a. Axial stress at the juncture of a flange and the hub of the flange.
- b. Thermal stresses.
- c. Membrane stress in the knuckle area of the head.
- d. Membrane stress due to local relenting loads.

Secondary bending stress, Q_b .

- a. Bending stress at a gross structural discontinuity: nozzles, lugs, etc. (relenting loadings only).
- b. The nonuniform portion of the stress distribution in a thick-walled vessel due to internal pressure.
- c. The stress variation of the radial stress due to internal pressure in thick-walled vessels.
- d. Discontinuity stresses at stiffening or support rings.

Note: For b and c it is necessary to subtract out the average stress which is the primary stress. Only the varying part of the stress distribution is a secondary stress.

Peak stress, F . Peak stresses are the additional stresses due to stress intensification in highly localized areas. They apply to both sustained loads and self-limiting loads. There are no significant distortions associated with peak stresses. Peak stresses are additive to primary and secondary stresses present at the point of the stress concentration. Peak stresses are only significant in fatigue conditions or brittle materials. Peak stresses are sources of fatigue cracks and apply to membrane, bending, and shear stresses. Examples are:

- a. Stress at the corner of a discontinuity.
- b. Thermal stresses in a wall caused by a sudden change in the surface temperature.
- c. Thermal stresses in cladding or weld overlay.
- d. Stress due to notch effect (stress concentration).

Categories of Stress

Once the various stresses of a component are calculated, they must be combined and this final result compared to an

allowable stress (see Table 1-1). The combined classes of stress due to a combination of loads acting at the same time are stress categories. Each category has assigned limits of stress based on the hazard it represents to the vessel. The following is derived basically from ASME Code, Section VIII, Division 2, simplified for application to Division 1 vessels and allowable stresses. It should be used as a guideline only because Division 1 recognizes only two categories of stress—primary membrane stress and primary bending stress. Since the calculations of most secondary (thermal and discontinuities) and peak stresses are not included in this book, these categories can be considered for reference only. In addition, Division 2 utilizes a factor K multiplied by the allowable stress for increase due to short-term loads due to seismic or upset conditions. It also sets allowable limits of combined stress for fatigue loading where secondary and peak stresses are major considerations. Table 1-1 sets allowable stresses for both stress classifications and stress categories.

Table 1-1
Allowable Stresses for Stress Classifications and Categories

Stress Classification or Category	Allowable Stress
General primary membrane, P_m	SE
General primary bending, P_b	$1.5SE < .9F_y$
Local primary membrane, P_L ($P_L = P_m + Q_{ms}$)	$1.5SE < .9F_y$
Secondary membrane, Q_m	$1.5SE < .9F_y$
Secondary bending, Q_b	$3SE < 2F_y < UTS$
Peak, F	$2S_a$
$P_m + P_b + Q_m^* + Q_b$	$3SE < 2F_y < UTS$
$P_L + P_b$	$1.5SE < .9F_y$
$P_m + P_b + Q_m^* + Q_b$	$3SE < 2F_y < UTS$
$P_m + P_b + Q_m^* + Q_b + F$	$2S_a$

Notes:

Q_{ms} = membrane stresses from sustained loads

Q_m^* = membrane stresses from relenting, self-limiting loads

S = allowable stress per ASME Code, Section VIII, Division 1, at design temperature

F_y = minimum specified yield strength at design temperature

UTS = minimum specified tensile strength

S_a = allowable stress for any given number of cycles from design fatigue curves.

SPECIAL PROBLEMS

This book provides detailed methods to cover those areas most frequently encountered in pressure vessel design. The topics chosen for this section, while of the utmost interest to the designer, represent problems of a specialized nature. As such, they are presented here for information purposes, and detailed solutions are not provided. The solutions to these special problems are complicated and normally beyond the expertise or available time of the average designer.

The designer should be familiar with these topics in order to recognize when special consideration is warranted. If more detailed information is desired, there is a great deal of reference material available, and special references have been included for this purpose. Whenever solutions to problems in any of these areas are required, the design or analysis should be referred to experts in the field who have proven experience in their solution.

Thick-Walled Pressure Vessels

As discussed previously, the equations used for design of thin-walled vessels are inadequate for design or prediction of failure of thick-walled vessels where $R_m/t < 10$. There are many types of vessels in the thick-walled vessel category as outlined in the following, but for purposes of discussion here only the monobloc type will be discussed. Design of thick-wall vessels or cylinders is beyond the scope of this book, but it is hoped that through the following discussion some insight will be gained.

In a thick-walled vessel subjected to internal pressure, both circumferential and radial stresses are maximum on the inside surface. However, failure of the shell does not begin at the bore but in fibers along the outside surface of the shell. Although the fibers on the inside surface do reach yield first they are incapable of failing because they are restricted by the outer portions of the shell. Above the elastic-breakdown pressure the region of plastic flow or "overstrain" moves radially outward and causes the circumferential stress to reduce at the inner layers and to increase at the outer layers. Thus the maximum hoop stress is reached first at the outside of the cylinder and eventual failure begins there.

The major methods for manufacture of thick-walled pressure vessels are as follows:

1. *Monobloc*—Solid vessel wall.
2. *Multilayer*—Begins with a core about $\frac{1}{2}$ in. thick and successive layers are applied. Each layer is vented (except the core) and welded individually with no overlapping welds.
3. *Multiwall*—Begins with a core about $1\frac{1}{2}$ in. to 2 in. thick. Outer layers about the same thickness are successively "shrunk fit" over the core. This creates compressive stress in the core, which is relaxed during pressurization. The process of compressing layers is called autofrettage from the French word meaning "self-hooping."
4. *Multilayer autofrettage*—Begins with a core about $\frac{1}{2}$ in. thick. Bands or forged rings are slipped outside

and then the core is expanded hydraulically. The core is stressed into plastic range but below ultimate strength. The outer rings are maintained at a margin below yield strength. The elastic deformation residual in the outer bands induces compressive stress in the core, which is relaxed during pressurization.

5. *Wire wrapped vessels*—Begin with inner core of thickness less than required for pressure. Core is wrapped with steel cables in tension until the desired autofrettage is achieved.
6. *Coil wrapped vessels*—Begin with a core that is subsequently wrapped or coiled with a thin steel sheet until the desired thickness is obtained. Only two longitudinal welds are used, one attaching the sheet to the core and the final closure weld. Vessels 5 to 6 ft in diameter for pressures up to 5,000 psi have been made in this manner.

Other techniques and variations of the foregoing have been used but these represent the major methods. Obviously these vessels are made for very high pressures and are very expensive.

For materials such as mild steel, which fail in shear rather than direct tension, the maximum shear theory of failure should be used. For internal pressure only, the maximum shear stress occurs on the inner surface of the cylinder. At this surface both tensile and compressive stresses are maximum. In a cylinder, the maximum tensile stress is the circumferential stress, σ_ϕ . The maximum compressive stress is the radial stress, σ_r . These stresses would be computed as follows:

$$\sigma_\phi = \frac{PR_i^2}{R_o^2 - R_i^2} \left(1 + \frac{R_o^2}{R_i^2} \right) = (+)$$

$$\sigma_r = \frac{PR_i^2}{R_o^2 - R_i^2} \left(1 - \frac{R_o^2}{R_i^2} \right) = (-)$$

Therefore the maximum shear stress, τ , is [9]:

$$\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{\sigma_\phi - \sigma_r}{2} = \frac{PR_o^2}{R_o^2 - R_i^2}$$

ASME Code, Section VIII, Division 1, has developed alternate equations for thick-walled monobloc vessels. The equations for thickness of cylindrical shells and spherical shells are as follows:

- *Cylindrical shells* (Para. 1-2 (a) (1)) where $t > .5 R_i$ or $P > .385 SE$:

$$Z = \frac{SE + P}{SE - P}$$

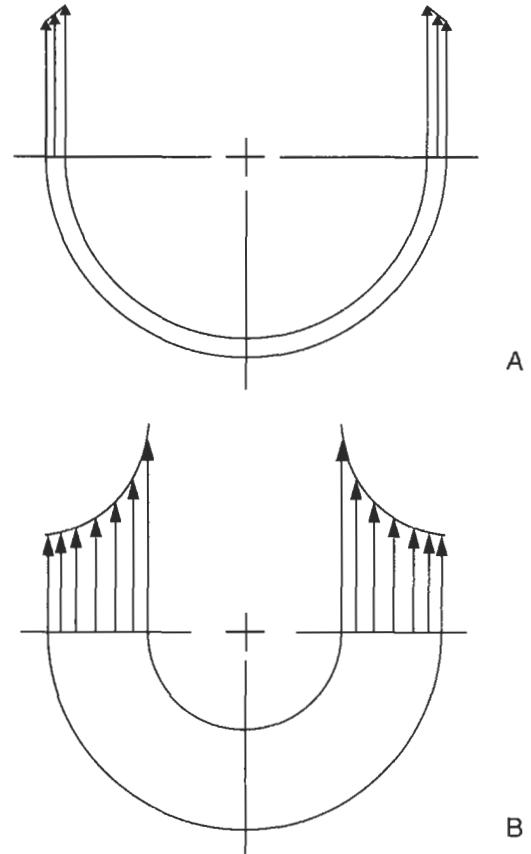


Figure 1-3. Comparison of stress distribution between thin-walled (A) and thick-walled (B) vessels.

$$t = \frac{R_o(\sqrt[3]{Z} - 1)}{Z}$$

- *Spherical shells* (Para. 1-3) where $t > .356 R_i$ or $P > .665 SE$:

$$Y = \frac{2(SE + P)}{2SE - P}$$

$$t = R_o \left(\frac{\sqrt[3]{Y} - 1}{\sqrt[3]{Y}} \right)$$

The stress distribution in the vessel wall of a thick-walled vessel varies across the section. This is also true for thin-walled vessels, but for purposes of analysis the stress is considered uniform since the difference between the inner and outer surface is slight. A visual comparison is offered in Figure 1-3.

Thermal Stresses

Whenever the expansion or contraction that would occur normally as a result of heating or cooling an object is prevented, thermal stresses are developed. The stress is always caused by some form of mechanical restraint.

Thermal stresses are "secondary stresses" because they are self-limiting. That is, yielding or deformation of the part relaxes the stress (except thermal stress ratcheting). Thermal stresses will not cause failure by rupture in ductile materials except by fatigue over repeated applications. They can, however, cause failure due to excessive deformations.

Mechanical restraints are either internal or external. External restraint occurs when an object or component is supported or contained in a manner that restricts thermal movement. An example of external restraint occurs when piping expands into a vessel nozzle creating a radial load on the vessel shell. Internal restraint occurs when the temperature through an object is not uniform. Stresses from a "thermal gradient" are due to internal restraint. Stress is caused by a thermal gradient whenever the temperature distribution or variation within a member creates a differential expansion such that the natural growth of one fiber is influenced by the different growth requirements of adjacent fibers. The result is distortion or warpage.

A transient thermal gradient occurs during heat-up and cool-down cycles where the thermal gradient is changing with time.

Thermal gradients can be logarithmic or linear across a vessel wall. Given a steady heat input inside or outside a tube the heat distribution will be logarithmic if there is a temperature difference between the inside and outside of the tube. This effect is significant for thick-walled vessels. A linear temperature distribution occurs if the wall is thin. Stress calculations are much simpler for linear distribution.

Thermal stress ratcheting is progressive incremental inelastic deformation or strain that occurs in a component that is subjected to variations of mechanical and thermal stress. Cyclic strain accumulation ultimately can lead to incremental collapse. Thermal stress ratcheting is the result of a sustained load and a cyclically applied temperature distribution.

The fundamental difference between mechanical stresses and thermal stresses lies in the nature of the loading. Thermal stresses as previously stated are a result of restraint or temperature distribution. The fibers at high temperature are compressed and those at lower temperatures are stretched. The stress pattern must only satisfy the requirements for equilibrium of the internal forces. The result being that yielding will relax the thermal stress. If a part is loaded mechanically beyond its yield strength, the part will continue to yield until it breaks, unless the deflection is limited by strain hardening or stress redistribution. The external load remains constant, thus the internal stresses cannot relax.

The basic equations for thermal stress are simple but become increasingly complex when subjected to variables such as thermal gradients, transient thermal gradients, logarithmic gradients, and partial restraint. The basic equations follow. If the temperature of a unit cube is changed

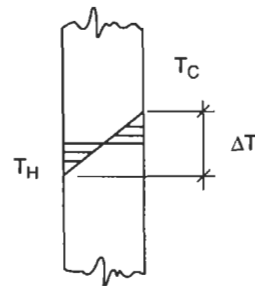


Figure 1-4. Thermal linear gradient across shell wall.

from T_1 to T_2 and the growth of the cube is fully restrained:

where T_1 = initial temperature, °F

T_2 = new temperature, °F

α = mean coefficient of thermal expansion in./in./°F

E = modulus of elasticity, psi

ν = Poisson's ratio = .3 for steel

ΔT = mean temperature difference, °F

Case 1: If the bar is restricted only in one direction but free to expand in the other direction, the resulting uniaxial stress, σ , would be

$$\sigma = -E\alpha(T_2 - T_1)$$

- If $T_2 > T_1$, σ is compressive (expansion).
- If $T_1 > T_2$, σ is tensile (contraction).

Case 2: If restraint is in both directions, x and y , then:

$$\sigma_x = \sigma_y = -\alpha E \Delta T / (1 - \nu)$$

Case 3: If restraint is in all three directions, x , y , and z , then

$$\sigma_x = \sigma_y = \sigma_z = -\alpha E \Delta T / (1 - 2\nu)$$

Case 4: If a thermal linear gradient is across the wall of a thin shell (see Figure 1-4), then:

$$\sigma_x = \sigma_\phi = \pm \alpha E \Delta T / 2(1 - \nu)$$

This is a bending stress and not a membrane stress. The hot side is in tension, the cold side in compression. Note that this is independent of vessel diameter or thickness. The stress is due to internal restraint.

Discontinuity Stresses

Vessel sections of different thickness, material, diameter, and change in directions would all have different displacements if allowed to expand freely. However, since they are connected in a continuous structure, they must deflect and rotate together. The stresses in the respective parts at or near the juncture are called discontinuity stresses. Discontinuity stresses are necessary to satisfy compatibility of deformation in the region. They are local in extent but can be of

very high magnitude. Discontinuity stresses are "secondary stresses" and are self-limiting. That is, once the structure has yielded, the stresses are reduced. In average application they will not lead to failure. Discontinuity stresses do become an important factor in fatigue design where cyclic loading is a consideration. Design of the juncture of the two parts is a major consideration in reducing discontinuity stresses.

In order to find the state of stress in a pressure vessel, it is necessary to find both the membrane stresses and the discontinuity stresses. From superposition of these two states of stress, the total stresses are obtained. Generally when combined, a higher allowable stress is permitted. Due to the complexity of determining discontinuity stress, solutions will not be covered in detail here. The designer should be aware that for designs of high pressure ($>1,500$ psi), brittle material or cyclic loading, discontinuity stresses may be a major consideration.

Since discontinuity stresses are self-limiting, allowable stresses can be very high. One example specifically addressed by the ASME Code, Section VIII, Division 1, is discontinuity stresses at cone-cylinder intersections where the included angle is greater than 60° . Para. 1-5(e) recommends limiting combined stresses (membrane + discontinuity) in the longitudinal direction to $4SE$ and in the circumferential direction to $1.5SE$.

ASME Code, Section VIII, Division 2, limits the combined stress, primary membrane and discontinuity stresses to $3S_m$, where S_m is the lesser of $\frac{2}{3}F_y$ or $\frac{1}{3}U.T.S.$, whichever is lower.

There are two major methods for determining discontinuity stresses:

1. *Displacement Method*—Conditions of equilibrium are expressed in terms of displacement.
2. *Force Method*—Conditions of compatibility of displacements are expressed in terms of forces.

See References 2, Article 4-7; 6, Chapter 8; and 7, Chapter 4 for detailed information regarding calculation of discontinuity stresses.

Fatigue Analysis

ASME Code, Section VIII, Division 1, does not specifically provide for design of vessels in cyclic service.

Although considered beyond the scope of this text as well, the designer must be aware of conditions that would require a fatigue analysis to be made.

When a vessel is subject to repeated loading that could cause failure by the development of a progressive fracture, the vessel is in cyclic service. ASME Code, Section VIII, Division 2, has established specific criteria for determining when a vessel must be designed for fatigue.

It is recognized that Code formulas for design of details, such as heads, can result in yielding in localized regions. Thus localized stresses exceeding the yield point may be encountered even though low allowable stresses have been used in the design. These vessels, while safe for relatively static conditions of loading, would develop "progressive fracture" after a large number of repeated loadings due to these high localized and secondary bending stresses. It should be noted that vessels in cyclic service require special consideration in both design and fabrication.

Fatigue failure can also be a result of thermal variations as well as other loadings. Fatigue failure has occurred in boiler drums due to temperature variations in the shell at the feed water inlet. In cases such as this, design details are of extreme importance.

Behavior of metal under fatigue conditions varies significantly from normal stress-strain relationships. Damage accumulates during each cycle of loading and develops at localized regions of high stress until subsequent repetitions finally cause visible cracks to grow, join, and spread. Design details play a major role in eliminating regions of stress raisers and discontinuities. It is not uncommon to have the design strength cut in half by poor design details. Progressive fractures develop from these discontinuities even though the stress is well below the static elastic strength of the material.

In fatigue service the localized stresses at abrupt changes in section, such as at a head junction or nozzle opening, misalignment, defects in construction, and thermal gradients are the significant stresses.

The determination of the need for a fatigue evaluation is in itself a complex job best left to those experienced in this type of analysis. For specific requirements for determining if a fatigue analysis is required see ASME Code, Section VIII, Division 2, Para. AD-160.

For additional information regarding designing pressure vessels for fatigue see Reference 7, Chapter 5.

REFERENCES

1. ASME Boiler and Pressure Vessel Code, Section VIII, Division 1, 1995 Edition, American Society of Mechanical Engineers.
2. ASME Boiler and Pressure Vessel Code, Section VIII, Division 2, 1995 Edition, American Society of Mechanical Engineers.
3. Popov, E. P., *Mechanics of Materials*, Prentice Hall, Inc., 1952.
4. Bednar, H. H., *Pressure Vessel Design Handbook*, Van Nostrand Reinhold Co., 1981.
5. Harvey, J. F., *Theory and Design of Modern Pressure Vessels*, Van Nostrand Reinhold Co., 1974.
6. Hicks, E. J. (Ed.), *Pressure Vessels—A Workbook for Engineers*, Pressure Vessel Workshop, Energy—Sources Technology Conference and Exhibition, Houston, American Society of Petroleum Engineers, January 19–21, 1981.
7. *Pressure Vessel and Piping Design*, Collected Papers 1927–1959, American Society of Mechanical Engineers, 1960.
8. Brownell, L. E., and Young, E. H., *Process Equipment Design*, John Wiley and Sons, 1959.
9. Roark, R. J., and Young, W. C., *Formulas for Stress and Strain*, 5th Edition, McGraw Hill Book Co., 1975.
10. Burgreen, D., *Design Methods for Power Plant Structures*, C. P. Press, 1975.
11. Criteria of the ASME Boiler and Pressure Vessel Code for Design by Analysis in Sections III and VIII, Division 2, American Society of Mechanical Engineers.

2 General Design

PROCEDURE 2-1

GENERAL VESSEL FORMULAS [1, 2]

Notation

- P = internal pressure, psi
 D_i, D_o = inside/outside diameter, in.
 S = allowable or calculated stress, psi
 E = joint efficiency
 L = crown radius, in.
 R_i, R_o = inside/outside radius, in.
 K, M = coefficients (See Note 3)
 σ_x = longitudinal stress, psi
 σ_ϕ = circumferential stress, psi
 R_m = mean radius of shell, in.
 t = thickness or thickness required of shell, head, or cone, in.
 r = knuckle radius, in.

Notes

- Formulas are valid for:
 - Pressures $< 3,000$ psi.
 - Cylindrical shells where $t \leq 0.5 R_i$ or $P \leq 0.385 SE$. For thicker shells see Reference 1, Para. 1-2.
 - Spherical shells and hemispherical heads where $t \leq 0.356 R_i$ or $P \leq 0.665 SE$. For thicker shells see Reference 1, Para. 1-3.
- All ellipsoidal and torispherical heads having a minimum specified tensile strength greater than 80,000 psi shall be designed using $S = 20,000$ psi at ambient temperature and reduced by the ratio of the allowable stresses at design temperature and ambient temperature where required.

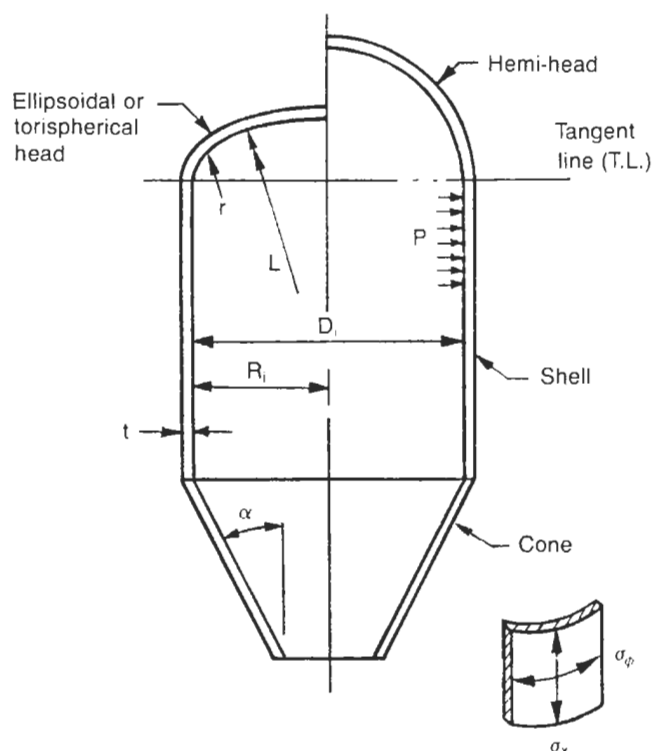


Figure 2-1. General configuration and dimensional data for vessel shells and heads.

- Formulas for factors:

$$K = 0.167 \left[2 + \left(\frac{D}{2h} \right)^2 \right]$$

$$M = 0.25 \left(3 + \sqrt{\frac{L}{r}} \right)$$

Table 2-1
General Vessel Formulas

Part	Stress Formula	Thickness, t		Pressure, P		Stress, S	
		I.D.	O.D.	I.D.	O.D.	I.D.	O.D.
Shell							
Longitudinal [1, Section UG-27(c)(2)]	$\sigma_x = \frac{PR_m}{0.2t}$	$\frac{PR_i}{2SE + 0.4P}$	$\frac{PR_o}{2SE + 1.4P}$	$\frac{2SE}{R_i - 0.4t}$	$\frac{2SE}{R_o - 1.4t}$	$\frac{P(R_i - 0.4t)}{2Et}$	$\frac{P(R_o - 1.4t)}{2Et}$
Circumferential [1, Section UG-27(c)(1); Section 1-1(a)(1)]	$\sigma_\phi = \frac{PR_m}{t}$	$\frac{PR_i}{SE - 0.6P}$	$\frac{PR_o}{SE + 0.4P}$	$\frac{SE}{R_i + 0.6t}$	$\frac{SE}{R_o - 0.4t}$	$\frac{P(R_i + 0.6t)}{Et}$	$\frac{P(R_o - 0.4t)}{Et}$
Heads							
Hemi sphere [1, Section 1-1(a)(2); Section UG-27(d)]	$\sigma_x = \sigma_\phi = \frac{PR_m}{2t}$	$\frac{PR_i}{2SE - 0.2P}$	$\frac{PR_o}{2SE + 0.8P}$	$\frac{2SE}{R_i + 0.2t}$	$\frac{2SE}{R_o - 0.8t}$	$\frac{P(R_i + 0.2t)}{2Et}$	$\frac{P(R_o - 0.8t)}{2Et}$
Ellipsoidal [1, Section 1-4(c)]	See Procedure 2-2	$\frac{PD_iK}{2SE - 0.2P}$	$\frac{PD_oK}{2SE + 2P(K - 0.1)}$	$\frac{2SE}{KD_i + 0.2t}$	$\frac{2SE}{KD_o - 2t(K - 0.1)}$	See Procedure 2-2	
2:1 S.E. [1, Section UG-32(d)]	"	$\frac{PD_i}{2SE - 0.2P}$	$\frac{PD_o}{2SE + 1.8P}$	$\frac{2SE}{D_i + 0.2t}$	$\frac{2SE}{D_o - 1.8t}$	"	
100%-6% Torispherical [1, Section UG-32(e)]	"	$\frac{0.885PL_i}{SE - 0.1P}$	$\frac{0.885PL_o}{SE + 0.8P}$	$\frac{SE}{0.885L_i + 0.1t}$	$\frac{SE}{0.885L_o - 0.8t}$	"	
Torispherical $L/r < 16.66$ [1, Section 1-4(d)]	"	$\frac{PL_iM}{2SE - 0.2P}$	$\frac{PL_oM}{2SE + P(M - 0.2)}$	$\frac{2SE}{L_iM + 0.2t}$	$\frac{2SE}{L_oM - t(M - 0.2)}$	"	
Cone							
Longitudinal	$\sigma_x = \frac{PR_m}{2t\cos \alpha}$	$\frac{PD_i}{4\cos \alpha (SE + 0.4P)}$	$\frac{PD_o}{4\cos \alpha (SE + 1.4P)}$	$\frac{4SE\cos \alpha}{D_i - 0.8t\cos \alpha}$	$\frac{4SE\cos \alpha}{D_o - 2.8t\cos \alpha}$	$\frac{P(D_i - 0.8t\cos \alpha)}{4Et\cos \alpha}$	$\frac{P(D_o - 2.8t\cos \alpha)}{4Et\cos \alpha}$
Circumferential [1, Section 1-4(e); Section UG-32(g)]	$\sigma_\phi = \frac{PR_m}{t\cos \alpha}$	$\frac{PD_i}{2\cos \alpha (SE - 0.6P)}$	$\frac{PD_o}{2\cos \alpha (SE + 0.4P)}$	$\frac{2SE\cos \alpha}{D_i + 1.2t\cos \alpha}$	$\frac{2SE\cos \alpha}{D_o - 0.8t\cos \alpha}$	$\frac{P(D_i + 1.2t\cos \alpha)}{2Et\cos \alpha}$	$\frac{P(D_o - 0.8t\cos \alpha)}{2Et\cos \alpha}$

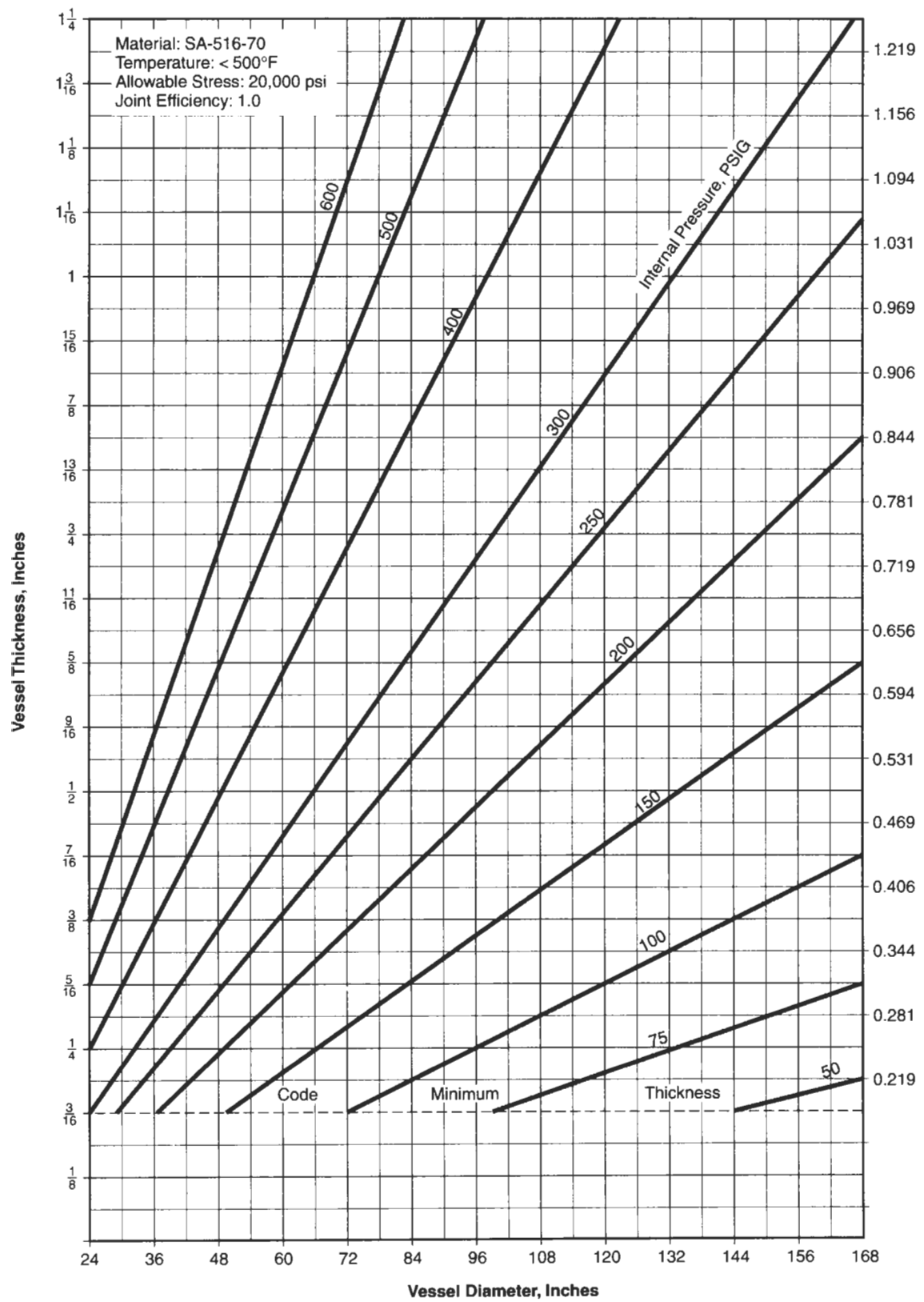


Figure 2-1a. Required shell thickness of cylindrical shell.

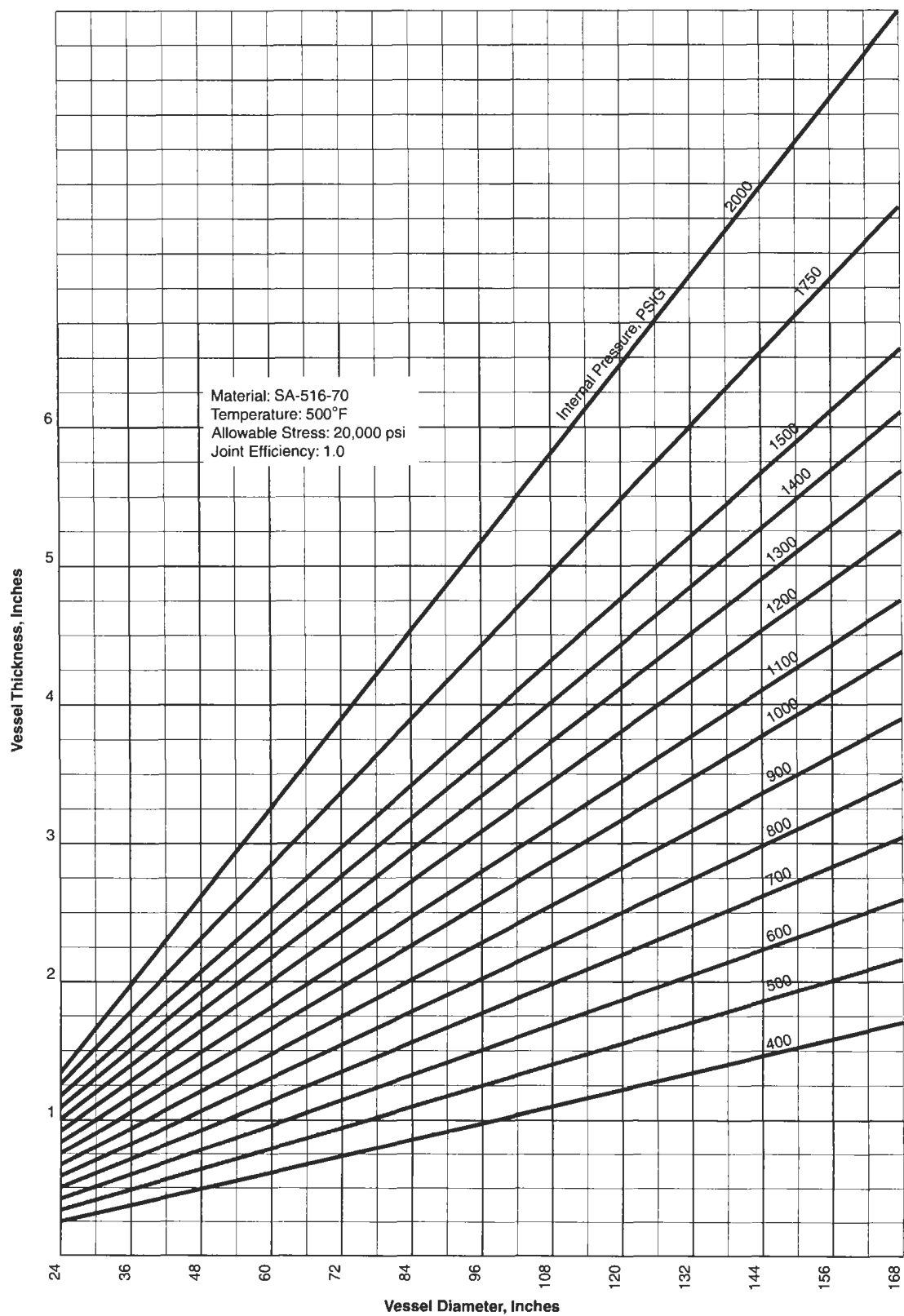


Figure 2-1a. (Continued)

PROCEDURE 2-2

EXTERNAL PRESSURE DESIGN

Notation

- A = factor "A," strain, from ASME Section II, Part D, Subpart 3, dimensionless
 A_s = cross-sectional area of stiffener, in.²
 B = factor "B," allowable compressive stress, from ASME Section II, Part D, Subpart 3, psi
 D = inside diameter of cylinder, in.
 D_o = outside diameter of cylinder, in.
 D_1 = outside diameter of the large end of cone, in.
 D_s = outside diameter of small end of cone, in.
 E = modulus of elasticity, psi
 I = actual moment of inertia of stiffener, in.⁴
 I_s = required moment of inertia of stiffener, in.⁴
 I'_s = required moment of inertia of combined shell-ring cross section, in.⁴
 L = for cylinders—the design length for external pressure, including $\frac{1}{3}$ the depth of heads, in.
 For cones—the design length for external pressure (see Figures 2-1b and 2-1c), in.
 L_c = equivalent length of conical section, in.
 L_s = length between stiffeners, in.
 $L_T - T$ = length of straight portion of shell, tangent to tangent, in.
 P = design internal pressure, psi
 P_a = allowable external pressure, psi
 P_x = design external pressure, psi
 R_o = outside radius of spheres and hemispheres, crown radius of torispherical heads, in.
 t = thickness of cylinder, head or conical section, in.
 t_c = equivalent thickness of cone, in.
 α = half apex angle of cone, degrees

Unlike vessels which are designed for internal pressure alone, there is no single formula, or unique design, which fits the external pressure condition. Instead, there is a range of options available to the designer which can satisfy the solution of the design. The thickness of the cylinder is only one part of the design. Other factors which affect the design are the length of cylinder and the use, size, and spacing of stiffening rings. Designing vessels for external pressure is an iterative procedure. First, a design is selected with all of the variables included, then the design is checked to determine if it is adequate. If inadequate, the procedure is repeated until an acceptable design is reached.

Vessels subject to external pressure may fail at well below the yield strength of the material. The geometry of the part is

the critical factor rather than material strength. Failures can occur suddenly, by collapse of the component.

External pressure can be caused in pressure vessels by a variety of conditions and circumstances. The design pressure may be less than atmospheric due to condensing gas or steam. Often refineries and chemical plants design all of their vessels for some amount of external pressure, regardless of the intended service, to allow for steam cleaning and the effects of the condensing steam. Other vessels are in vacuum service by nature of venturi devices or connection to a vacuum pump. Vacuums can be pulled inadvertently by failure to vent a vessel during draining, or from improperly sized vents.

External pressure can also be created when vessels are jacketed or when components are within multichambered vessels. Often these conditions can be many times greater than atmospheric pressure.

When vessels are designed for both internal and external pressure, it is common practice to first determine the shell thickness required for the internal pressure condition, then check that thickness for the maximum allowable external pressure. If the design is not adequate then a decision is made to either bump up the shell thickness to the next thickness of plate available, or add stiffening rings to reduce the "L" dimension. If the option of adding stiffening rings is selected, then the spacing can be determined to suit the vessel configuration.

Neither increasing the shell thickness to remove stiffening rings nor using the thinnest shell with the maximum number of stiffeners is economical. The optimum solution lies somewhere between these two extremes. Typically, the utilization of rings with a spacing of $2D$ for vessel diameters up to about eight feet in diameter and a ring spacing of approximately "D" for diameters greater than eight feet, provides an economical solution.

The design of the stiffeners themselves is also a trial and error procedure. The first trial will be quite close if the old API-ASME formula is used. The formula is as follows:

$$I_s = \frac{0.16D_o^3 P_x L_s}{E}$$

Stiffeners should never be located over circumferential weld seams. If properly spaced they may also double as insulation support rings. Vacuum stiffeners, if combined with other stiffening rings, such as cone reinforcement rings or saddle stiffeners on horizontal vessels, must be designed for the combined condition, not each independently. If at all

possible, stiffeners should always clear shell nozzles. If unavoidable, special attention should be given to the design of a boxed stiffener or connection to the nozzle neck.

Design Procedure For Cylindrical Shells

Step 1: Assume a thickness if one is not already determined.

Step 2: Calculate dimensions "L" and "D." Dimension "L" should include one-third the depth of the heads. The overall length of cylinder would be as follows for the various head types:

$$\begin{array}{ll} W/(2) \text{ hemi-heads} & L = L_{T-T} + 0.333D \\ W/(2) \text{ 2:1 S.E. heads} & L = L_{T-T} + 0.1666D \\ W/(2) \text{ 100\% - 6\% heads} & L = L_{T-T} + 0.112D \end{array}$$

Step 3: Calculate L/D_o and D_o/t ratios

Step 4: Determine Factor "A" from ASME Code, Section II, Part D, Subpart 3, Fig G: Geometric Chart for Components Under External or Compressive Loadings (see Figure 2-1e).

Step 5: Using Factor "A" determined in step 4, enter the applicable material chart from ASME Code, Section II, Part D, Subpart 3 at the appropriate temperature and determine Factor "B."

Step 6: If Factor "A" falls to the left of the material line, then utilize the following equation to determine the allowable external pressure:

$$P_a = \frac{2AE}{3(D_o/t)}$$

Step 7: For values of "A" falling on the material line of the applicable material chart, the allowable external pressure should be computed as follows:

$$P_a = \frac{4B}{3(D_o/t)}$$

Step 8: If the computed allowable external pressure is less than the design external pressure, then a decision must be made on how to proceed. Either (a) select a new thickness and start the procedure from the beginning or (b) elect to use stiffening rings to reduce the "L" dimension. If stiffening rings are to be utilized, then proceed with the following steps.

Step 9: Select a stiffener spacing based on the maximum length of unstiffened shell (see Table 2-1a). The stiffener spacing can vary up to the maximum value allowable for the assumed thickness. Determine the number of stiffeners necessary and the corresponding "L" dimension.

Step 10: Assume an approximate ring size based on the following equation:

$$I = \frac{0.16D_o^3 P_x L_s}{E}$$

Step 11: Compute Factor "B" from the following equation utilizing the area of the ring selected:

$$B = \frac{0.75PD_o}{t + A_s/L_s}$$

Step 12: Utilizing Factor "B" computed in step 11, find the corresponding "A" Factor from the applicable material curve.

Step 13: Determine the required moment of inertia from the following equation. Note that Factor "A" is the one found in step 12.

$$I_s = \frac{[D_o^2 L_s (t + A_s/L_s) A]}{14}$$

Step 14: Compare the required moment of inertia, I , with the actual moment of inertia of the selected member. If the actual exceeds that which is required, the design is acceptable but may not be optimum. The optimization process is an iterative process in which a new member is selected, and steps 11 through 13 are repeated until the required size and actual size are approximately equal.

Notes

1. For conical sections where $\alpha < 22.5$ degrees, design the cone as a cylinder where $D_o = D_L$ and length is equal to L .
2. If a vessel is designed for less than 15 psi, and the external pressure condition is not going to be stamped on the nameplate, the vessel does not have to be designed for the external pressure condition.

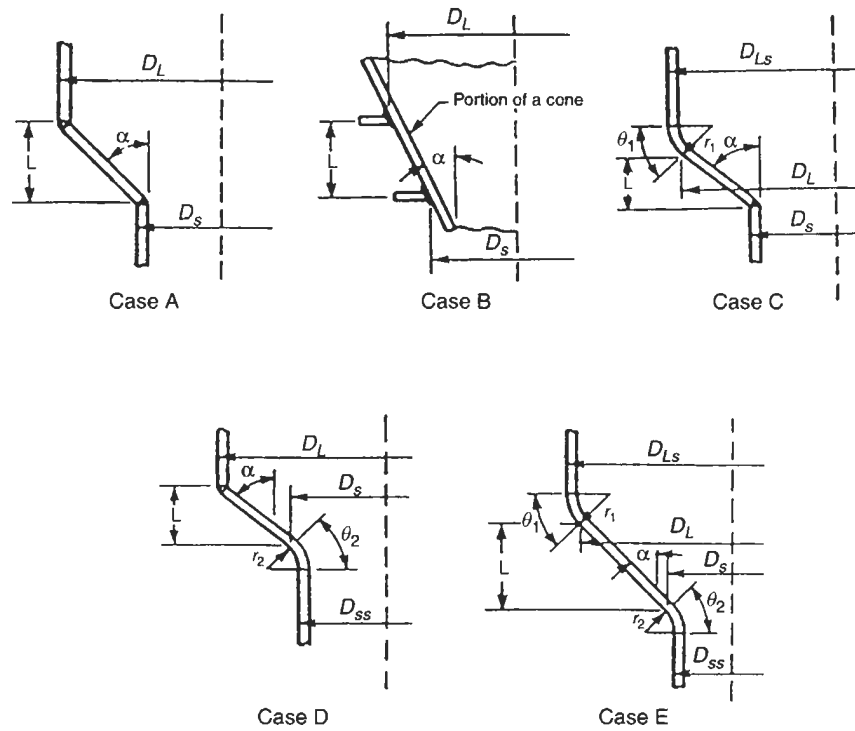


Figure 2-1b. External pressure cones $22\frac{1}{2}^\circ < \alpha < 60^\circ$.

For Case B, $L_c = L$

For Cases A, C, D, E:

$$L_c = 0.5 \left(1 + \frac{D_s}{D_L} \right)$$

$$t_c = t \cos \alpha$$

$$L_c/D_L =$$

$$D_L/t_c =$$

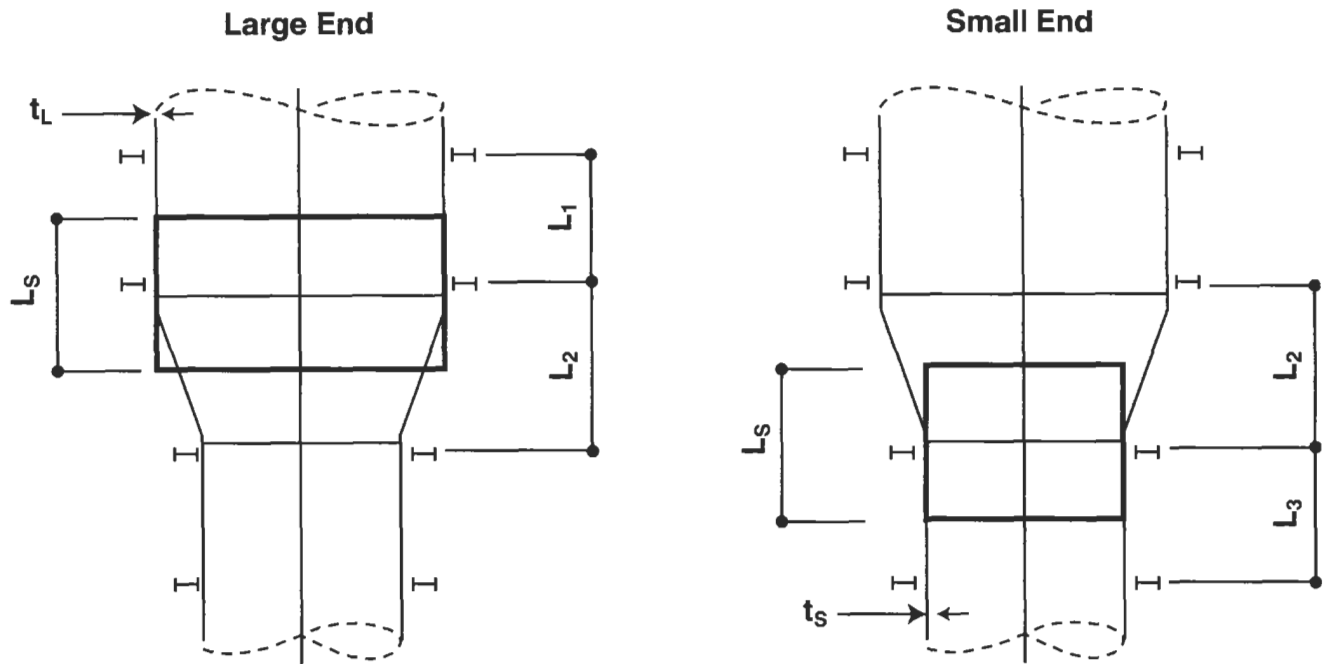


Figure 2-1c. Combined shell/cone design for stiffened shells.

Design stiffener for large end of cone as cylinder where:

$$D_o = D_L$$

$$t = t_L$$

$$L_s = \frac{L_1}{2} + \frac{L_2}{2}$$

Design stiffener for small end of cone as cylinder where:

$$D_o = D_s$$

$$t = t_s$$

$$L_s = \frac{L_3}{2} + \frac{L_2}{2} \left[1 + \frac{D_s}{D_L} \right]$$

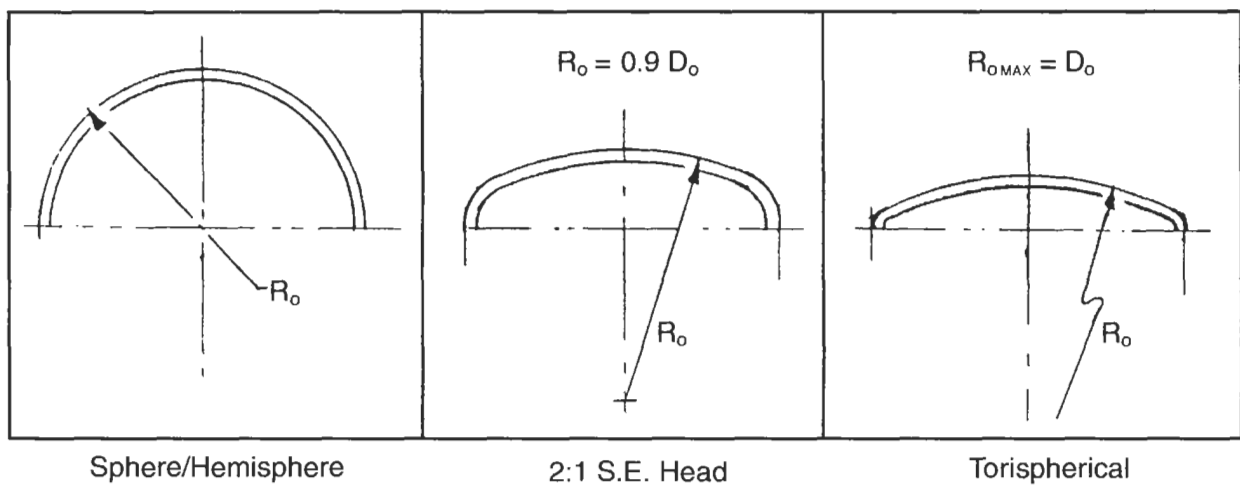


Figure 2-1d. External pressure ~ spheres and heads.

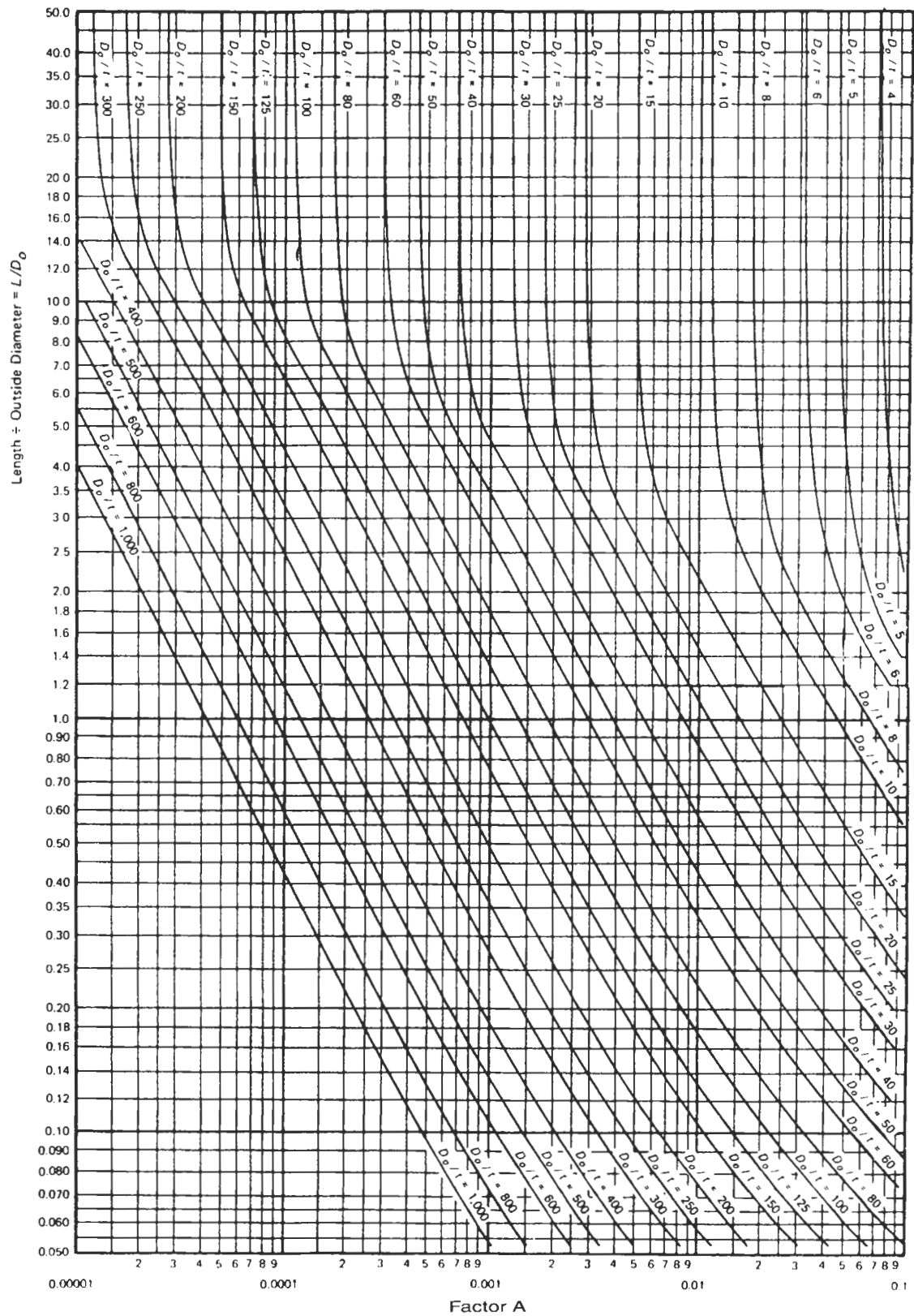


Figure 2-1e. Geometric chart for components under external or compressive loadings (for all materials). (Reprinted by permission from the ASME Code, Section VIII, Div. 1.)

Design Procedure For Spheres and Heads

Step 1: Assume a thickness and calculate Factor "A."

$$A = \frac{0.125t}{R_o}$$

Step 2: Find Factor "B" from applicable material chart.

B =

Step 3: Compute P_a .

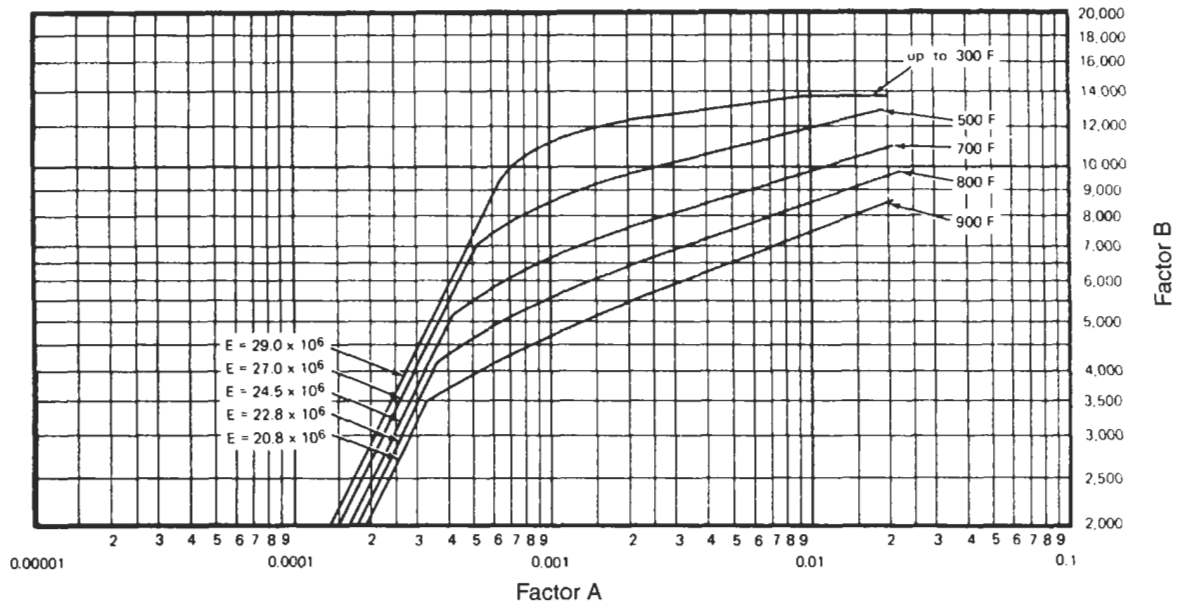


Figure 2-1f. Chart for determining shell thickness of components under external pressure when constructed of carbon or low-alloy steels (specified minimum yield strength 24,000 psi to, but not including, 30,000 psi). (Reprinted by permission from the ASME Code, Section VIII, Div. 1.)

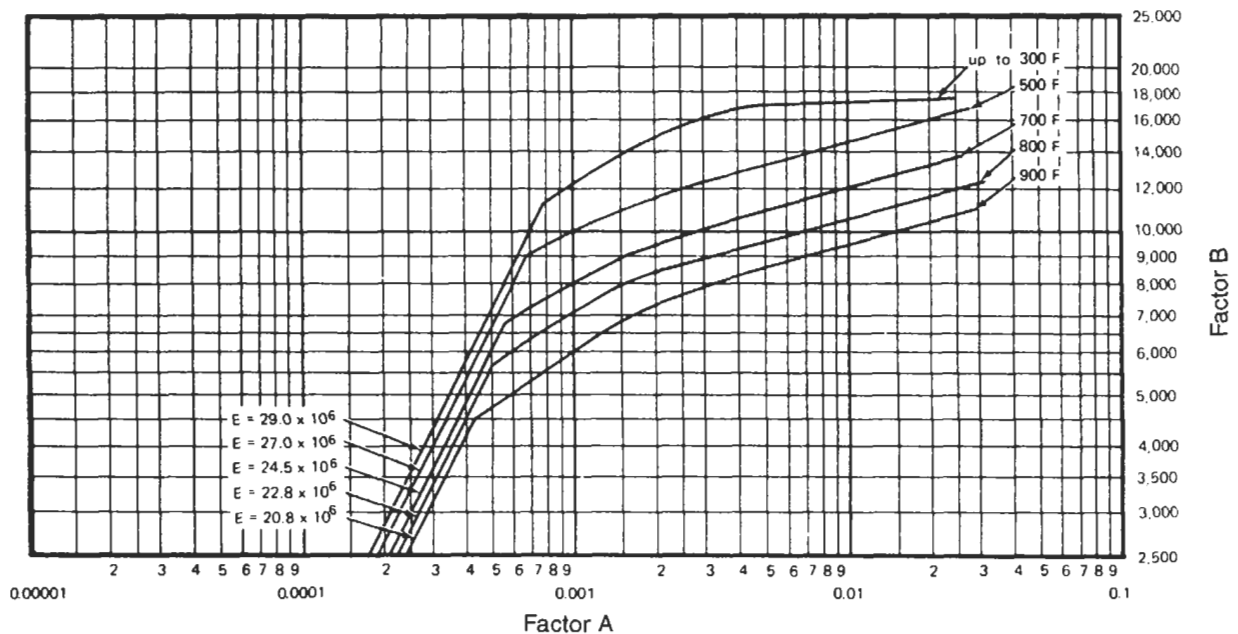


Figure 2-1g. Chart for determining shell thickness of components under external pressure when constructed of carbon or low-alloy steels (specified minimum yield strength 30,000 psi and over except materials within this range where other specific charts are referenced) and type 405 and type 410 stainless steels. (Reprinted by permission from the ASME Code, Section VIII, Div. 1.)

$$\text{A to left of material line} \quad P_a = \frac{0.0625E}{(R_o/t)^2}$$

$$\text{A to right of material line} \quad P_a = \frac{Bt}{R_o}$$

Notes

1. As an alternative, the thickness required for 2:1 S.E. heads for external pressure may be computed from the formula for internal pressure where $P = 1.67 P_x$ and $E = 1.0$.

Table 2-1a
Maximum Length of Unstiffened Shells

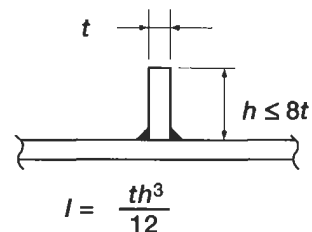
Diameter (in.)	Thickness (in.)															
	1/4	5/16	3/8	7/16	1/2	9/16	5/8	11/16	3/4	13/16	7/8	15/16	1	1 1/16	1 1/8	1 3/16
36	204															
	∞															
42	168	280														
	313	∞														
48	142	235	358													
	264	437	∞													
54	122	203	306	437												
	228	377	∞													
60	104	178	268	381												
	200	330	499	∞												
66	91	157	238	336	458											
	174	293	442	626	∞											
72	79	138	213	302	408	537										
	152	263	396	561	∞											
78	70	124	193	273	369	483	616									
	136	237	359	508	686	∞										
84	63	110	175	249	336	438	559									
	123	212	327	462	625	816	∞									
90	57	99	157	228	308	402	510	637								
	112	190	300	424	573	748	∞									
96	52	90	143	210	284	370	470	585	715							
	103	173	274	391	528	689	875	∞								
102	48	82	130	190	263	343	435	540	661	795						
	94	160	249	363	490	639	810	1,005	∞							
108	44	76	118	176	245	320	405	502	613	738	875					
	87	148	228	337	456	594	754	935	∞							
114	42	70	109	162	223	299	379	469	571	687	816					
	79	138	211	311	426	555	705	874	1,064	∞						
120	39	65	101	149	209	280	355	440	536	642	762	894				
	74	128	197	287	400	521	660	819	997	∞						
126	37	61	95	138	195	263	334	414	504	603	715	839	974			
	69	120	184	266	374	490	621	770	938	1,124	∞					
132	35	57	88	129	181	242	315	391	475	569	673	789	916	1,053		
	65	113	173	248	348	462	586	727	884	1,060	1,253	∞				
138	33	54	83	121	169	228	297	369	449	538	636	744	864	994		
	62	106	163	234	325	437	555	687	836	1,002	1,185	∞				
144	31	51	78	114	158	214	275	350	426	510	603	705	817	940	1,073	
	59	98	154	221	304	411	526	652	793	950	1,123	1,312	∞			
150		49	74	107	148	201	261	332	405	485	573	669	774	891	1,017	1,152
		92	146	209	286	385	499	619	753	902	1,066	1,246	1,442	∞		
156		46	70	101	140	189	248	309	385	462	546	637	737	846	966	1,095
		87	138	199	271	363	475	590	717	859	1,015	1,186	1,373	∞		
162		44	67	96	133	178	233	294	367	440	520	608	703	806	919	1,042
		83	131	189	258	342	448	562	684	819	968	1,131	1,309	1,509	∞	
	1/4	5/16	3/8	7/16	1/2	9/16	5/8	11/16	3/4	13/16	7/8	15/16	1	1 1/16	1 1/8	1 3/16

Notes:

1. All values are in in.
2. Values are for temperatures up to 500°F.
3. Top value is for full vacuum, lower value is half vacuum.
4. Values are for carbon or low-alloy steel ($F_y > 30,000$ psi) based on Figure 2-1g.

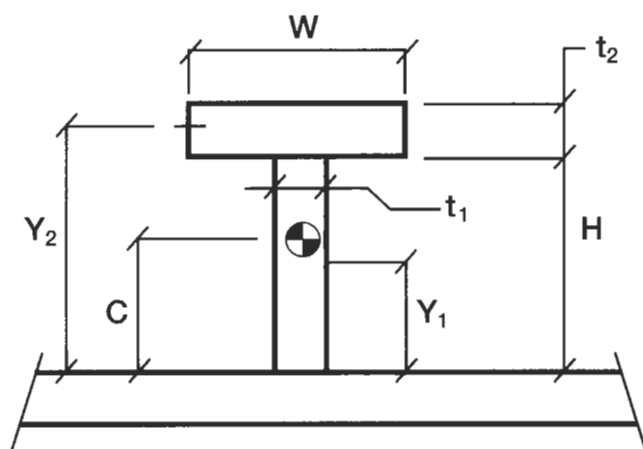
Table 2-1b
Moment of Inertia of Bar Stiffeners

Thk t, in.	Max, ht, in.	Height, h, in.														
		1	1½	2	2½	3	3½	4	4½	5	5½	6	6½	7	7½	8
¼	2	0.020 0.250	0.070 0.375	0.167 0.5												
5/16	2.5	0.026 0.313	0.088 0.469	0.208 0.625	0.407 0.781											
3/8	3	0.031 0.375	0.105 0.563	0.25 0.75	0.488 0.938	0.844 1.125										
7/16	3.5		0.123 0.656	0.292 0.875	0.570 1.094	0.984 1.313	1.563 1.531									
½	4		0.141 0.75	0.333 1.00	0.651 1.25	1.125 1.50	1.786 1.75	2.667 2.00								
9/16	4.5			0.375 1.125	0.732 1.406	1.266 1.688	2.00 1.969	3.00 2.25	4.271 2.53							
5/8	5				0.814 1.563	1.41 1.875	2.23 2.188	3.33 2.50	4.75 2.813	6.510 3.125						
11/16	5.5					1.55 2.063	2.46 2.406	3.67 2.75	5.22 3.094	7.16 3.438	9.53 3.78					
¾	6					1.69 2.25	2.68 2.625	4.00 3.00	5.70 3.375	7.81 3.75	10.40 4.125	13.5 4.50				
13/16	6.5						2.90 2.844	4.33 3.25	6.17 3.656	8.46 4.063	11.26 4.469	14.63 4.875	18.59 5.281			
7/8	7							4.67 3.50	6.64 3.94	9.11 4.375	12.13 4.813	15.75 5.25	20.02 5.688	25.01 6.125		
1	8							5.33 4.00	7.59 4.50	10.42 5.00	13.86 5.50	18.00 6.00	22.89 6.50	28.58 7.00	35.16 7.50	42.67 8.00



Note: Upper value in table is the moment of inertia. Lower value is the area.

Table 2-1c
Moment of Inertia of Composite Stiffeners



$$I_1 = \frac{t_1 H^3}{12}$$

$$I_2 = \frac{W t_2^3}{12}$$

$$c = \frac{\sum A_n Y_n}{\sum A}$$

$$I = \sum A_n Y_n^2 + \sum I - C \sum A_n Y_n$$

Type	H	W	t ₁	t ₂	Σ A	Σ I	C	I
1	3	3	0.375	0.5	2.63	0.87	2.50	2.84
2	3	4	0.5	0.5	3.50	1.17	2.50	-3.80
3	4	4	0.375	0.5	3.50	2.04	3.28	6.45
4	4	5	0.5	0.625	5.13	2.77	3.41	9.28
5	4.5	5	0.5	0.5	4.75	3.85	3.57	0.125
6	5	4	0.5	0.625	5.00	5.29	3.91	15.12
7	5.5	4	0.5	0.5	4.75	6.97	4.01	17.39
8	6	5	0.5	0.625	6.13	9.10	4.69	25.92
9	6	6	0.625	0.625	7.50	11.37	4.66	31.82
10	5.5	6	0.875	0.875	10.01	12.47	4.42	37.98
11	6.5	6	0.75	0.75	9.38	17.37	4.99	48.14
12	7	6	0.625	0.75	8.88	18.07	5.46	51.60
13	8	6	0.75	1	12.00	32.50	6.25	93.25
14	8	6	1	1	14.00	43.16	5.93	112.47

Moment of Inertia of Stiffening Rings

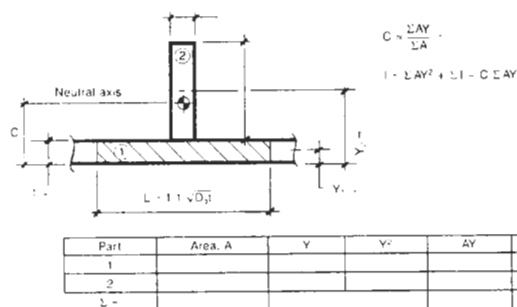


Figure 2-1h. Case 1: Bar-type stiffening ring.

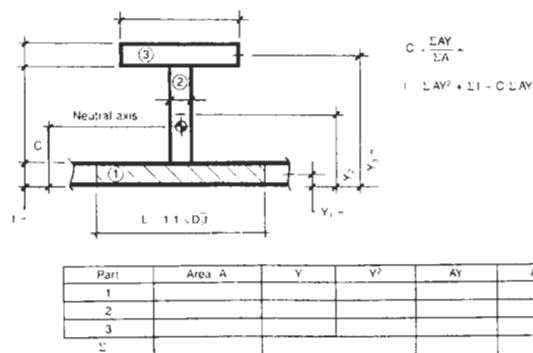


Figure 2-1i. Case 2: T-type stiffening ring.

STIFFENING RING CHECK FOR EXTERNAL PRESSURE

Ls		$B = 0.75 \frac{PD_o}{t + A_s/L_s}$		Moment of inertia w/o shell	
t				$I_s = \frac{D_o^2 L_s (t + A_s/L_s) A}{14}$	
P		If $B \leq 2,500$ psi, $A = 2B/E$		Moment of inertia w/ shell	
D _o					
A _s		If $B > 2,500$ psi, determine A from applicable material charts			
E = modulus of elasticity					

From Ref. 1, Section UG-29.

PROCEDURE 2-3

CALCULATE MAP, MAWP, AND TEST PRESSURES

Notation

S_a = allowable stress at ambient temperature, psi
 S_{DT} = allowable stress at design temperature, psi
 S_{CA} = allowable stress of clad material at ambient temperature, psi
 S_{CD} = allowable stress of clad material at design temperature, psi
 S_{BA} = allowable stress of base material at ambient temperature, psi
 S_{BD} = allowable stress of base material at design temperature, psi
 $C.a.$ = corrosion allowance, in.
 t_{sc} = thickness of shell, corroded, in.
 t_{sn} = thickness of shell, new, in.
 t_{hc} = thickness of head, corroded, in.
 t_{hn} = thickness of head, new, in.
 t_b = thickness of base portion of clad material, in.
 t_c = thickness of cladding, in.
 R_n = inside radius, new, in.
 R_c = inside radius, corroded, in.
 R_o = outside radius, in.
 D_n = inside diameter, new, in.
 D_c = inside diameter, corroded, in.
 D_o = outside diameter, in.
 P_M = MAP, psi
 P_W = MAWP, psi
 P = design pressure, psi
 P_s = shop hydro pressure (new and cold), psi
 P_F = field hydro pressure (hot and corroded), psi

E = joint efficiency, see Procedure 2-1
 and Appendix C

Definitions

Maximum Allowable Working Pressure (MAWP): The MAWP for a vessel is the maximum permissible pressure at the top of the vessel in its normal operating position at a specific temperature, usually the design temperature. When calculated, the MAWP should be stamped on the nameplate. The MAWP is the maximum pressure allowable in the "hot and corroded" condition. It is the least of the values calculated for the MAWP of any of the essential parts of the vessel, and adjusted for any difference in static head that may exist between the part considered and the top of the vessel. This pressure is based on calculations for every element of the vessel using nominal thicknesses exclusive of corrosion allowance. It is the basis for establishing the set pressures of any pressure-relieving devices protecting the vessel. The design pressure may be substituted if the MAWP is not calculated.

The MAWP for any vessel part is the maximum internal or external pressure, including any static head, together with the effect of any combination of loadings listed in UG-22 which are likely to occur, exclusive of corrosion allowance at the designated coincident operating temperature. The MAWP for the vessel will be governed by the MAWP of the weakest part.

The MAWP may be determined for more than one designated operating temperature. The applicable allowable

stress value at each temperature would be used. When more than one set of conditions is specified for a given vessel, the vessel designer and user should decide which set of conditions will govern for the setting of the relief valve.

Maximum Allowable Pressure (MAP): The term MAP is often used. It refers to the maximum permissible pressure based on the weakest part in the new (uncorroded) and cold condition, and all other loadings are not taken into consideration.

Design Pressure: The pressure used in the design of a vessel component for the most severe condition of coincident pressure and temperature expected in normal operation. For this condition, and test condition, the maximum difference in pressure between the inside and outside of a vessel, or between any two chambers of a combination unit, shall be considered. Any thickness required for static head or other loadings shall be additional to that required for the design pressure.

Design Temperature: For most vessels, it is the temperature that corresponds to the design pressure. However, there is a maximum design temperature and a minimum design temperature for any given vessel. The minimum design temperature would be the MDMT (see Procedure 2-17). The MDMT shall be the lowest temperature expected in service or the lowest allowable temperature as calculated for the individual parts. Design temperature for vessels under external pressure shall not exceed the maximum temperatures given on the external pressure charts.

Operating Pressure: The pressure at the top of the vessel at which it normally operates. It shall be lower than the MAWP, design pressure, or the set pressure of any pressure relieving device.

Operating Temperature: The temperature that will be maintained in the metal of the part of the vessel being considered for the specified operation of the vessel.

Calculations

- MAWP, corroded at Design Temperature P_w .

Shell:

$$P_w = \frac{S_{DT} E t_{sc}}{R_c + 0.6 t_{sc}} \text{ or } \frac{S_{DT} E t_{sc}}{R_o - 0.4 t_{sc}}$$

2:1 S.E. Head:

$$P_w = \frac{2 S_{DT} E t_{hc}}{D_c + 0.2 t_{hc}} \text{ or } \frac{2 S_{DT} E t_{hc}}{D_o - 1.8 t_{hc}}$$

- MAP, new and cold, P_M .

Shell:

$$P_M = \frac{S_a E t_{sn}}{R_n + 0.6 t_{sn}} \text{ or } \frac{S_a E t_{sn}}{R_o - 0.4 t_{sn}}$$

2:1 S.E. Head:

$$P_M = \frac{2 S_a E t_{hn}}{D_n + 0.2 t_{hn}} \text{ or } \frac{2 S_a E t_{hn}}{D_o - 1.8 t_{hn}}$$

- Shop test pressure, P_s .

$$P_s = 1.3 P_M \text{ or } 1.3 P_w \left[\frac{S_a}{S_{DT}} \right]$$

- Field test pressure, P_F .

$$P_F = 1.3 P$$

- For clad vessels where credit is taken for the clad material, the following thicknesses may be substituted into the equations for MAP and MAWP:

$$t_{sc}, t_{hc} = t_b + \left[\frac{S_{CD}}{S_{BD}} (t_c - C.a.) \right]$$

$$t_{sn}, t_{hn} = t_b + \left[\frac{S_{CA} t_c}{S_{BA}} \right]$$

Notes

1. Also check the pressure-temperature rating of the flanges for MAWP and MAP.
2. All nozzles should be reinforced for MAWP.
3. The MAP and MAWP for other components, i.e., cones, flat heads, hemi-heads, torispherical heads, etc., may be checked in the same manner by using the formula for pressure found in Procedure 2-1 and substituting the appropriate terms into the equations.
4. It is not necessary to take credit for the cladding thickness. If it is elected not to take credit for the cladding thickness, then base all calculations on the full base metal thickness. This assumes the cladding is equivalent to a corrosion allowance, and no credit is taken for the strength of the cladding.

PROCEDURE 2-4

STRESSES IN HEADS DUE TO INTERNAL PRESSURE [2, 3]

Notation

L = crown radius, in.
 r = knuckle radius, in.
 h = depth of head, in.
 R_L = latitudinal radius of curvature, in.
 R_m = meridional radius of curvature, in.
 σ_ϕ = latitudinal stress, psi
 σ_x = meridional stress, psi
 P = internal pressure, psi

Formulas

Lengths of R_L and R_m for ellipsoidal heads:

- At equator:

$$R_m = \frac{h^2}{R}$$

$$R_L = R$$

- At center of head:

$$R_m = R_L = \frac{R^2}{h}$$

- At any point X:

$$R_L = \sqrt{\frac{R^4}{h^2} + X^2 \left(1 - \frac{R^2}{h^2}\right)}$$

$$R_m = \frac{R_L^3 h^2}{R^4}$$

Notes

1. Latitudinal (hoop) stresses in the knuckle become compressive when the R/h ratio exceeds 1.42. These heads will fail by either elastic or plastic buckling, depending on the R/t ratio.
2. Head types fall into one of three general categories: hemispherical, torispherical, and ellipsoidal. Hemispherical heads are analyzed as spheres and were

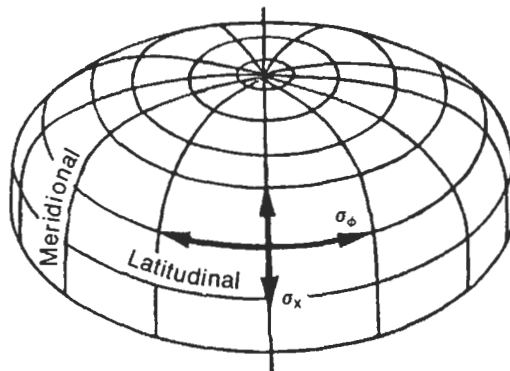


Figure 2-2. Direction of stresses in a vessel head.

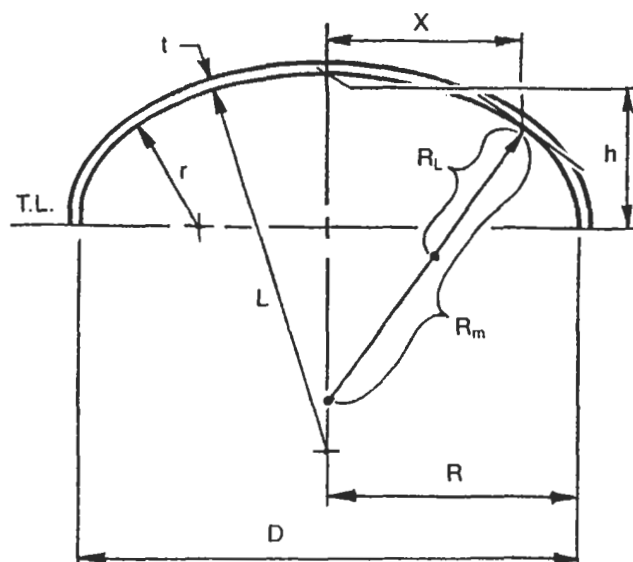


Figure 2-3. Dimensional data for a vessel head.

covered in the previous section. Torispherical (also known as flanged and dished heads) and ellipsoidal head formulas for stress are outlined in the following form.

TORISPHERICAL HEADS			
σ_x		σ_ϕ	
At Junction of Crown and Knuckle			
$\sigma_x = \frac{PL}{2t}$		$\sigma_\phi = \frac{PL}{4t} \left(3 - \frac{L}{R} \right)$	
In Crown			
$\sigma_x = \frac{PL}{2t}$		$\sigma_\phi = \sigma_x$	
In Knuckle			
$\sigma_x = \frac{PL}{2t}$		$\sigma_\phi = \frac{PL}{t} \left(1 - \frac{L}{2r} \right)$	
At Tangent Line			
$\sigma_x = \frac{PR}{2t}$		$\sigma_\phi = \frac{PR}{t}$	

ELLIPSOIDAL HEADS			
σ_x		σ_ϕ	
At Any Point X			
$\sigma_x = \frac{PR_L}{2t}$		$\sigma_\phi = \frac{PR_L}{t} \left(1 - \frac{R_L}{2R_m} \right)$	
At Center of Head			
$\sigma_x = \frac{PR^2}{2th}$		$\sigma_\phi = \sigma_x$	
At Tangent Line			
$\sigma_x = \frac{PR}{2t}$		$\sigma_\phi = \frac{PR}{t} \left(1 - \frac{R^2}{2h^2} \right)$	

PROCEDURE 2-5

DESIGN OF INTERMEDIATE HEADS [1, 3]

Notation

A = factor A for external pressure
 A_s = shear area, in.²
 B = allowable compressive stress, psi
 F = load on weld(s), lb/in.
 τ = shear stress, psi
 E = joint efficiency
 E_1 = modulus of elasticity at temperature, psi
 S = code allowable stress, psi
 H_D = hydrostatic end force, lb
 P_i = maximum differential pressure on concave side of head, psi

P_e = maximum differential pressure on convex side of head, psi

K = spherical radius factor (see Table 2-2)

L = inside radius of hemi-head, in.

= 0.9D for 2:1 S.E. heads, in.

= KD for ellipsoidal heads, in.

= crown radius of F & D heads, in.

Table 2-2
 Spherical Radius Factor, K

$\frac{D}{2h}$	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0
K	.5	.57	.65	.73	.81	.9	.99	1.08	1.18	1.27	1.36

Reprinted by permission from ASME Code Section VIII Div. 1, Table UG-33.1.

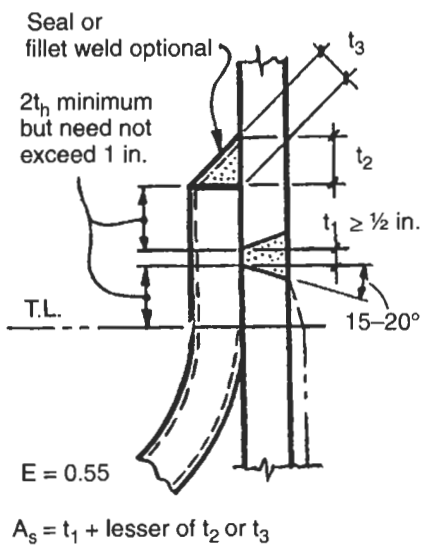


Figure 2-4. Dimensional data for an intermediate head.

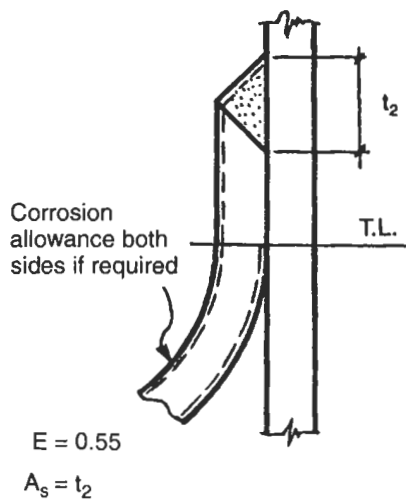
Required Head Thickness, t_r

- *Internal pressure, P_i .* Select appropriate head formula based on head geometry. For dished only heads as in Figure 2-5, Case 3:

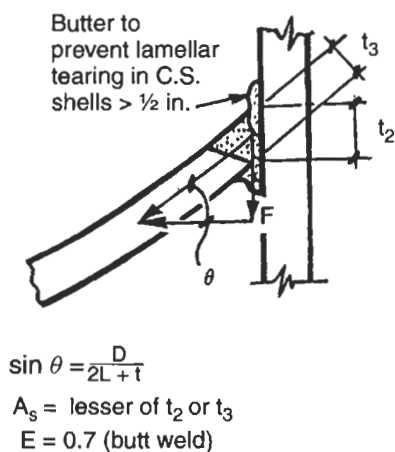
$$t_r = \frac{5P_i L}{6S}$$



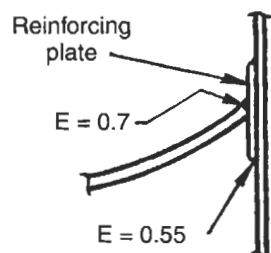
Case 1



Case 2



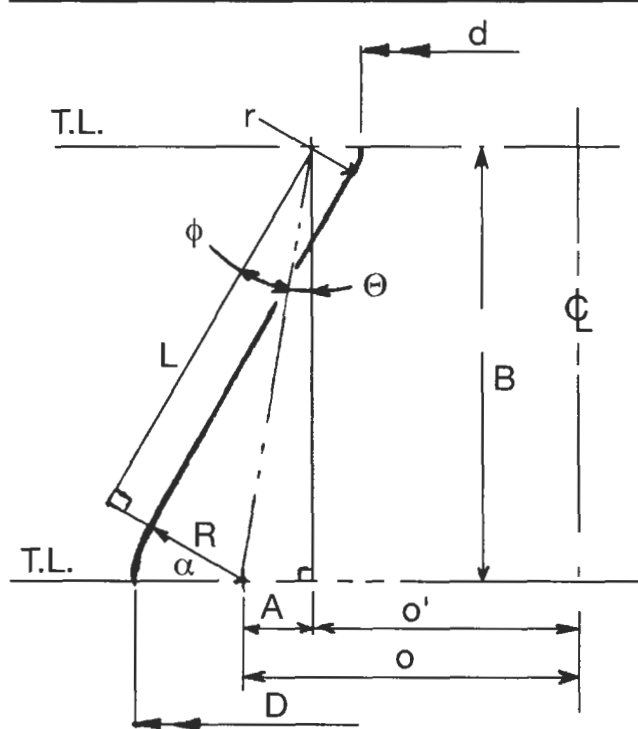
Case 3



Design the weld attaching the head as in Case 3 and the welds attaching the reinforcing plate to share full load

Case 3 Alternate

Figure 2-5. Methods of attachment of intermediate heads.

Calculating Angle, α 

Case 1

$$o > o'$$

Step 1:

$$\sin \phi = \frac{R + r}{\sqrt{A^2 + B^2}} =$$

$$\phi =$$

Step 2:

$$\tan \Theta = \frac{A}{B}$$

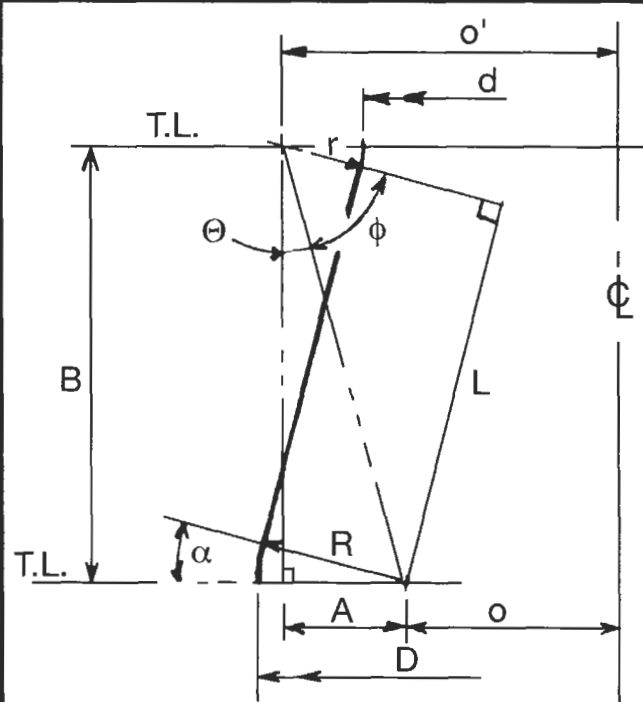
$$\Theta =$$

Step 3:

$$\alpha = \phi + \Theta$$

$$\alpha =$$

$$L = \cos \phi \sqrt{A^2 + B^2}$$



Case 2

$$o' > o$$

Step 1:

$$\cos \phi = \frac{R + r}{\sqrt{A^2 + B^2}} =$$

$$\phi =$$

Step 2:

$$\tan \Theta = \frac{A}{B}$$

$$\Theta =$$

Step 3:

$$\alpha = 90 - \Theta - \phi$$

$$\alpha =$$

$$L = \sin \phi \sqrt{A^2 + B^2}$$

Dimensional Formulas

$$D_1 = D - 2(R - R \cos \alpha)$$

$$D_2 = D + 2(R - R \cos \alpha)$$

$$D' = D - 2R \left(1 - \frac{1}{\cos \alpha} \right) - 2\ell \tan \alpha$$

$$L_1 = \frac{D_1}{2 \cos \alpha}$$

$$L_2 = \frac{D_2}{2 \cos \alpha}$$

$$m = 0.25 \left(3 + \sqrt{\frac{L_1}{R}} \right)$$

Large End (Figure 2-7)

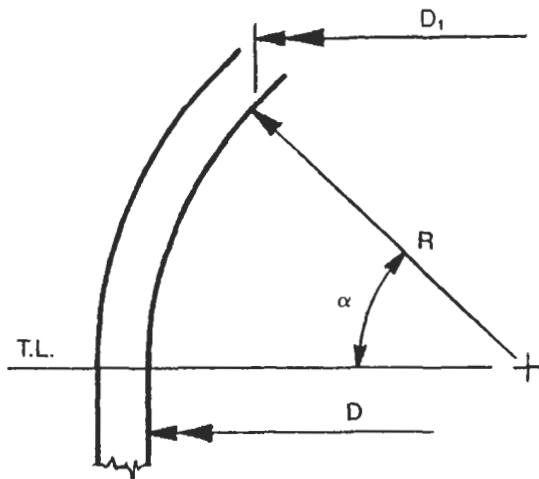


Figure 2-7. Dimensional data for the large end of a conical transition.

- Maximum longitudinal loads, f_1 .

(+) tension; (-) compression

$$f_1 = \frac{-W_1}{\pi D_1} \pm \frac{4M_1}{\pi D_1^2}$$

- Determine equivalent pressure, P_1 .

$$P_1 = P + \frac{4f_1}{D_1}$$

- Circumferential stress, D_1 .

Compression:

$$\sigma_1 = \frac{PL_1}{t} - \frac{P_1 L_1}{t} \left[\frac{L_1}{2R} \right]$$

- Circumferential stress at D_1 without loads, σ_1 .

Compression:

$$\sigma_1 = \frac{PL_1}{t} \left(1 - \frac{L_1}{2R} \right)$$

- Thickness required knuckle, t_{rk} [1, section 1-4(d)].

With loads:

$$t_{rk} = \frac{P_1 L_1 m}{2SE - 0.2P_1}$$

Without loads:

$$t_{rk} = \frac{PL_1 m}{2SE - 0.2P}$$

- Thickness required cone, t_{rc} [1, section UG-32(g)].

With loads:

$$t_{rc} = \frac{P_1 D_1}{2 \cos \alpha (SE - 0.6P_1)}$$

Without loads:

$$t_{rc} = \frac{PD_1}{2 \cos \alpha (SE - 0.6P)}$$

Small End (Figure 2-8)

- Maximum longitudinal loads, f_2 .

(+) tension; (-) compression

$$f_2 = \frac{-W_2}{\pi D_2} \pm \frac{4M_2}{\pi D_2^2}$$

- Determine equivalent pressure, P_2 .

$$P_2 = P + \frac{4f_2}{D_2}$$

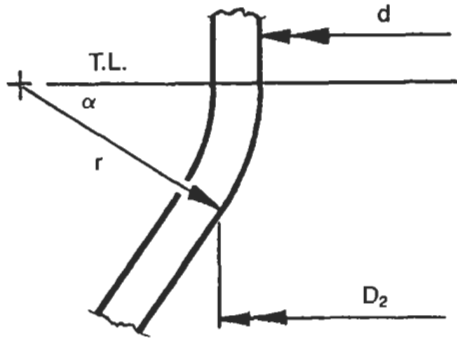


Figure 2-8. Dimensional data for the small end of a conical transition.

- Circumferential stress at D_2 .

Compression:

$$\sigma_2 = \frac{PL_2}{t} + \frac{P_2L_2}{t} \left[\frac{L_2}{2r} \right]$$

- Circumferential stress at D_2 without loads, σ_2 .

Compression:

$$\sigma_2 = \frac{PL_2}{t} \left(1 - \frac{L_2}{2r} \right)$$

- Thickness required cone, at D_2 , t_{rc} [1, section UG-32(g)].

With loads:

$$t_{rc} = \frac{P_2D_2}{2 \cos \alpha (SE - 0.6P_2)}$$

Without loads:

$$t_{rc} = \frac{PD_2}{2 \cos \alpha (SE - 0.6P)}$$

- Thickness required knuckle. There is no requirement for thickness of the reverse knuckle at the small end of the cone. For convenience of fabrication it should be made the same thickness as the cone.

Additional Formulas (Figure 2-9)

- Thickness required of cone at any diameter D' , $t_{D'}$.

$$t_{D'} = \frac{PD'}{2 \cos \alpha (SE - 0.6P)}$$

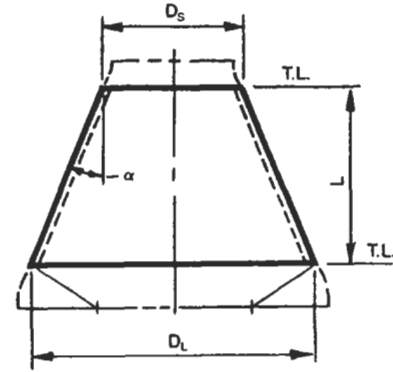


Figure 2-9. Dimensional data for cones due to external pressure.

- Thickness required for external pressure [1, section UG-33(f)].

$$t_e = t \cos \alpha$$

$$D_L = D_2 + 2t_e$$

$$D_s = D_1 + 2t_e$$

$$L = X - \sin \alpha (R + t) - \sin \alpha (r - t)$$

$$L_e = \frac{L}{2} \left(1 + \frac{D_s}{D_L} \right)$$

$$\frac{L_e}{D_L} =$$

$$\frac{D_L}{t_e} =$$

Using these values, use Figure 2-1e to determine Factor A.

- Allowable external pressure, P_a .

$$P_a = \frac{2AEt_e}{D_L}; \quad P_a > P_x$$

where E = modulus of elasticity at design temperature.

Notes

1. Allowable stresses. The maximum stress is the compressive stress at the tangency of the large knuckle and the cone. Failure would occur in local yielding rather than buckling; therefore the allowable stress should be the same as required for cylinders. Thus the allowable circumferential compressive stress should be the lesser of $2SE$ or F_y . Using a lower allowable stress would require the knuckle radius to be made very large—well above code requirements. See Reference 3.
2. Toriconical sections are mandatory if angle α exceeds 30° unless the design complies with Para. 1-5(e) of the

ASME Code [1]. This paragraph requires a discontinuity analysis of the cone-shell juncture.

3. No reinforcing rings or added reinforcement is required at the intersections of cones and cylinders, providing a knuckle radius meeting ASME Code requirements is used. The minimum knuckle radius for the large end is not less than the greater of $3t$ or $0.12(R + t)$. The knuckle radius of the small end (flare) has no minimum. (See [Reference 1, Figure UG-36]).

4. Toriconical transitions are advisable to avoid the high discontinuity stresses at the junctures for the following conditions:
 - a. High pressure—greater than 300 psig.
 - b. High temperature—greater than 450 or 500°F.
 - c. Low temperature—less than -20°F .
 - d. Cyclic service (fatigue).

PROCEDURE 2-7

DESIGN OF FLANGES [1, 4]

Notation

A = flange O.D., in.
 A_b = cross-sectional area of bolts, in.^2
 A_m = total required cross-sectional area of bolts, in.^2
 a = nominal bolt diameter, in.
 B = flange I.D., in. (see Note 6)
 B_1 = flange I.D., in. (see Note 6)
 B_s = bolt spacing, in.
 b = effective gasket width, in.
 b_o = gasket seating width, in.
 C = bolt circle diameter, in.
 d = hub shape factor
 d_1 = bolt hole diameter, in.
 E, h_D, h_G, h_T, R = radial distances, in.
 e = hub shape factor
 F = hub shape factor for integral-type flanges
 F_L = hub shape factor for loose-type flanges
 f = hub stress correction factor for integral flanges
 G = diameter at gasket load reaction, in.
 g_o = thickness of hub at small end, in.
 g_1 = thickness of hub at back of flange, in.
 H = hydrostatic end force, lb
 H_D = hydrostatic end force on area inside of flange, lb
 H_G = gasket load, operating, lb
 H_p = total joint-contact surface compression load, lb
 H_T = pressure force on flange face, lb
 h = hub length, in.
 h_o = hub factor
 M_D = moment due to H_D , in.-lb

M_G = moment due to H_G , in.-lb
 M_o = total moment on flange, operating, in.-lb
 M'_o = total moment on flange, seating
 M_T = moment due to H_T , in.-lb
 m = gasket factor (see Table 2-3)
 m_o = unit load, operating, lb
 m_g = unit load, gasket seating, lb
 N = width of gasket, in. (see Table 2-4)
 w = width of raised face or gasket contact width, in. (see Table 2-4)
 n = number of bolts
 ν = Poisson's ratio, 0.3 for steel
 P = design pressure, psi
 S_a = allowable stress, bolt, at ambient temperature, psi
 S_b = allowable stress, bolt, at design temperature, psi
 S_{fa} = allowable stress, flange, at ambient temperature, psi
 S_{fo} = allowable stress, flange, at design temperature, psi
 S_H = longitudinal hub stress, psi
 S_R = radial stress in flange, psi
 S_T = tangential stress in flange, psi T, U, Y
 Z = K-factors (see Table 2-5)
 T_r, U_r, Y_r = K-factors for reverse flanges
 t = flange thickness, in.
 t_n = pipe wall thickness, in.
 V = hub shape factor for integral flanges
 V_L = hub shape factor for loose flanges
 W = flange design bolt load, lb
 W_{m1} = required bolt load, operating, lb
 W_{m2} = required bolt load, gasket seating, lb
 y = gasket design seating stress, psi

Formulas

$$h_D = \frac{C - \text{dia. } H_D}{2}$$

$$h_T = \frac{C - \text{dia. } H_T}{2}$$

$$h_G = \frac{C - G}{2}$$

$$h_o = \sqrt{B g_o}$$

$$H_D = \frac{\pi B^2 P}{4}$$

$$H_T = H - H_D$$

$$H_G = \text{operating} = W_{m1} - H$$

$$\text{gasket seating} = W$$

$$H = \frac{G^2 \pi P}{4}$$

$$m_o = \frac{M_o}{B}$$

$$m_G = \frac{M_G}{B}$$

$$M_D = H_D h_D$$

$$M_T = H_T h_T$$

$$M_G = W h_G$$

$$E = \frac{A - C}{2}$$

$$K = \frac{A}{B}$$

$$T = \frac{(1 - \nu^2)(K^2 - 1)U}{(1 - \nu) + (1 + \nu)K^2}$$

$$Z = \frac{K^2 + 1}{K^2 - 1}$$

$$Y = (1 - \nu^2)U$$

$$U = \frac{K^2(1 + 4.6052(1 + \nu/1 - \nu)\log_{10} K) - 1}{1.0472(K^2 - 1)(K - 1)(1 + \nu)}$$

$$B_1 = \text{loose flanges} = B + g_1$$

$$= \text{integral flanges, } f < 1 = B + g_1$$

$$= \text{integral flanges, } f \geq 1 = B + g_o$$

$$d = \text{loose flanges} = \frac{U h_o g_o^2}{V_L}$$

$$= \text{integral flanges} = \frac{U h_o g_o^2}{V}$$

$$= \text{reverse flanges} = \frac{U_r h_o g_o^2}{V}$$

$$e = \text{loose flanges} = \frac{F_L}{h_o}$$

$$= \text{integral flanges} = \frac{F}{h_o}$$

$$G = (\text{if } b_o \leq 0.25 \text{ in.}) \text{ mean diameter of gasket face}$$

$$= (\text{if } b_o > 0.25 \text{ in.}) \text{ O.D. of gasket contact face} - 2b$$

Stress Formula Factors

$$\alpha = t e + 1$$

$$\beta = 1.333 t e + 1$$

$$\delta = \frac{t^3}{d}$$

$$\gamma = \frac{\alpha}{T} \text{ or } \frac{\alpha}{T_r} \text{ for reverse flanges}$$

$$\lambda = \gamma + \delta$$

$$\alpha_R = \frac{1}{K^2} \left[1 + \frac{3(K+1)(1-\nu)}{\pi Y} \right]$$

For factors, F , U , F_L , and U_L , see Table 2-7.1 of the ASME Code [1].

Special Flanges

Special flanges that are required to be designed should only be used as a last resort. Whenever possible, standard flanges should be utilized. In general, special designs as outlined in this procedure are done for large or high-pressure designs. Flanges in this category will be governed by one of two conditions:

1. Gasket seating force, W_{m2}
2. Hydrostatic end force, H

For high-pressure flanges, typically the hydrostatic end force, H , will govern. For low-pressure flanges, the gasket seating force will govern. Therefore the strategy for approaching the design of these flanges will vary. The strategy is as follows:

- *For low-pressure flanges*
 - a. Minimize the gasket width to reduce the force necessary to seat the gasket.
 - b. Use a larger number of smaller diameter bolts to minimize the bolt circle diameter and thus reduce the moment arm which governs the flange thickness.
 - c. Utilize hubless flanges (either lap joint or plate flanges) to minimize the cost of forgings.
- *For high-pressure flanges*
High-pressure flanges require a large bolt area to counteract the large hydrostatic end force. Large bolts, in turn, increase the bolt circle with a corresponding increase in the moment arm. Thicker flanges and large hubs are necessary to distribute the bolt loads. Seek a balance between the quantity and size of bolts, bolt spacing, and bolt circle diameter.

Design Strategy

Step 1: Determine the number and size of bolts required. As a rule of thumb, start with a number of bolts equal to the nominal size of the bore in inches, rounded to the nearest multiple of four. First, calculate W_{m1} or W_{m2} . A_m is equal to the larger of W_{m1} or W_{m2} divided by S_u . The quantity of

bolts required is:

$$n = A_m / R_u$$

To find the size of bolt for a given quantity:

$$R_u = A_m / n$$

With these two equations a variety of combinations can be determined.

Step 2: Determine the bolt circle diameter for the selected bolt size.

$$C = B + 2g + 2R$$

The flange O.D. may now be established.

$$A = C + 2E$$

Step 3: Check the minimum bolt spacing (not an ASME requirement). Compare with the value of B_s in Table 2-5a.

$$B_s = C/n$$

Note: Dimensions R_u , R , E , and B_s are from Table 2-5a.

Step 4: After all of the preliminary dimensions and details are selected, proceed with the detailed analysis of the flange by calculating the balance of forces, moments, and stresses in the appropriate design form.

Gasket Facing and Selection

The gasket facing and type correspond to the service conditions, fluid or gas handled, pressure, temperature, thermal shock, cyclic operation, and the gasket selection. The greater the hazard, the more care that should be invested in the decisions regarding gasket selection and facing details.

Facings which confine the gasket, male and female, tongue and groove and ring joint offer greater security against blowouts. Male and female and tongue and groove have the disadvantage that mating flanges are not alike. These facings, which confine the gasket, are known as enclosed gaskets and are required for certain services, such as TEMA Class "R."

For tongue and groove flanges, the tongue is more likely to be damaged than the groove; therefore, from a maintenance standpoint, there is an advantage in placing the tongue on the part which can be transported for servicing, i.e., blind flanges, manway heads, etc. If the assembly of these joints is horizontal then there will be less difficulty if the groove is placed in the lower side of the joint. The gasket width should be made equal to the width of the tongue. Gaskets for these joints are typically metal or metal jacketed.

TYPE 1: WELD NECK FLANGE DESIGN (INTEGRAL)

1 DESIGN CONDITIONS					
Design pressure, P			Allowable Stresses		
Design temperature			Flange		Bolting
Flange material			Design temp., S_{fo}		Design temp., S_b
Bolting material			Atm. temp., S_{fa}		Atm. temp., S_a
Corrosion allowance					
2 GASKET AND FACING DETAILS					
Gasket			Facing		
3 TABLES 2-3 AND 2-4			4 LOAD AND BOLT CALCULATIONS		
N			$W_{m2} = b\pi G y$	$A_m = \text{greater of } W_{m2}/S_a \text{ or } W_{m1}/S_b$	
b			$H_p = 2b\pi G m P$		
G			$H = G^2 \pi P / 4$	A_b	
y			$W_{m1} = H_p + H$	$W = 0.5(A_m + A_b)S_a$	
m					
5 MOMENT CALCULATIONS					
Load		×	Lever Arm		= Moment
Operating					
$H_D = \pi B^2 P / 4$			$h_D = R + 0.5g_1$	$M_D = H_D h_D$	
$H_G = W_{m1} - H$			$h_G = 0.5(C - G)$	$M_G = H_G h_G$	
$H_T = H - H_D$			$h_T = 0.5(R + g_1 + h_G)$	$M_T = H_T h_T$	
				M_o	
Seating					
$H_G = W$			$h_G = 0.5(C - G)$	M'_o	
6 K AND HUB FACTORS					
$K = A/B$			h/h_o		
T			F		
Z			V		
Y			f		
U			$e = F/h_o$		
g_1/g_o			$d = \frac{U}{V} h_o g_o^2$		
$h_o = \sqrt{B g_o}$					
7 STRESS FORMULA FACTORS					
t			$\alpha = t e + 1$		
$\beta = 4/3 t e + 1$					
$\gamma = \alpha/T$					
$\delta = t^3/d$					
$\lambda = \gamma + \delta$					
$m_o = M_o/B$					
$m_G = M'_o/B$					
If bolt spacing exceeds $2a + t$, multiply m_o and m_G in above equation by:			$\sqrt{\frac{\text{Bolt spacing}}{2a + t}}$		
8 STRESS CALCULATIONS					
Allowable Stress	Operating		Allowable Stress	Seating	
$1.5 S_{fo}$	Longitudinal hub, $S_H = f m_o \lambda g_1^2$		$1.5 S_{fa}$	Longitudinal hub, $S_H = f m_G \lambda g_1^2$	
S_{fo}	Radial flange, $S_R = \beta m_o / \lambda t^2$		S_{fa}	Radial flange, $S_R = \beta m_G / \lambda t^2$	
S_{fo}	Tangential flange, $S_T = m_o Y / t^2 - Z S_R$		S_{fa}	Tangential flange, $S_T = m_G Y / t^2 - Z S_R$	
S_{fo}	Greater of $0.5(S_H + S_R)$ or $0.5(S_H + S_T)$		S_{fa}	Greater of $0.5(S_H + S_R)$ or $0.5(S_H + S_T)$	

Figure 2-10. Dimensional data and forces for a weld neck flange (integral).

Adapted from Taylor Forge International, Inc., by permission.

TYPE 2: SLIP-ON FLANGE DESIGN (LOOSE)

1 DESIGN CONDITIONS					
Design pressure, P			Allowable Stresses		
Design temperature			Flange	Bolting	
Flange material		Design temp., S_{10}		Design temp., S_b	
Bolting material		Atm. temp., S_{1a}		Atm. temp., S_a	
Corrosion allowance					
2 GASKET AND FACING DETAILS					
Gasket			Facing		
3 TABLES 2-3 AND 2-4			4 LOAD AND BOLT CALCULATIONS		
N		$W_{m2} = b\pi Gy$		$A_m = \text{greater of } W_{m2}/S_a \text{ or } W_{m1}/S_b$	
b		$H_p = 2b\pi GmP$		A_b	
G		$H = G^2\pi P/4$		$W = 0.5(A_m + A_b)S_a$	
y		$W_{m1} = H_p + H$			
m					
5 MOMENT CALCULATIONS					
Load		x	Lever Arm		= Moment
Operating					
$H_D = \pi B^2 P/4$		$h_D = R + g_1$		$M_D = H_D h_D$	
$H_G = W_{m1} - H$		$h_G = 0.5(C - G)$		$M_G = H_G h_G$	
$H_T = H - H_D$		$h_T = 0.5(R + g_1 + h_G)$		$M_T = H_T h_T$	
				M_o	
Seating					
$H_G = W$		$h_G = 0.5(C - G)$		M'_o	
6 K AND HUB FACTORS					
$K = A/B$		h/h_o			
T		F_L			
Z		V_L			
Y					
U		$e = \frac{F_L}{h_o}$			
g_1/g_o		$d = \frac{U}{V} h_o g_o^2$			
$h_o = \sqrt{B g_o}$					
7 STRESS FORMULA FACTORS					
t					
$\alpha = te + 1$					
$\beta = 4/3 te + 1$					
$\gamma = \alpha/T$					
$\delta = t^2/d$					
$\lambda = \gamma + \delta$					
$m_o = M_o/B$					
$m_G = M'_o/B$					
If bolt spacing exceeds $2a + t$, multiply m_o and m_g in above equation by: $\sqrt{\frac{\text{Bolt spacing}}{2a + t}}$					
8 STRESS CALCULATIONS					
Allowable Stress	Operating		Allowable Stress	Seating	
$1.5 S_{10}$	Longitudinal hub, $S_H = m_o/\lambda g_1^2$		$1.5 S_{1a}$	Longitudinal hub, $S_H = m_o/\lambda g_1^2$	
S_{10}	Radial flange, $S_R = \beta m_o/\lambda t^2$		S_{1a}	Radial flange, $S_R = \beta m_o/\lambda t^2$	
S_{10}	Tangential flange, $S_T = m_o Y/t^2 - Z S_R$		S_{1a}	Tangential flange, $S_T = m_o Y/t^2 - Z S_R$	
S_{10}	Greater of $0.5(S_H + S_R)$ or $0.5(S_H + S_T)$		S_{1a}	Greater of $0.5(S_H + S_R)$ or $0.5(S_H + S_T)$	

Figure 2-11. Dimensional data and forces for a slip-on flange (loose).

Adapted from Taylor Forge International, Inc., by permission.

TYPE 3: RING FLANGE DESIGN

1 DESIGN CONDITIONS				
Design pressure, P		Allowable Stresses		
Design temperature		Flange	Bolting	
Flange material		Design temp., S_{10}	Design temp., S_b	
Bolting material		Atm. temp., S_{1a}	Atm. temp., S_a	
Corrosion allowance				
2 GASKET AND FACING DETAILS				
Gasket		Facing		
3 TABLES 2-3 AND 2-4		4 LOAD AND BOLT CALCULATIONS		
N		$W_{m2} = b \pi G y$	$A_m = \text{greater of}$	
b		$H_p = 2b \pi G m P$	$W_{m2}/S_a \text{ or } W_{m1}/S_b$	
G		$H = G^2 \pi P/4$	A_b	
y		$W_{m1} = H_p + H$	$W = 0.5(A_m + A_b)/S_a$	
m				
5 MOMENT CALCULATIONS				
Load	x	Lever Arm	=	Moment
Operating				
$H_D = \pi B^2 P/4$		$h_D = 0.5(C - B)$		$M_D = H_D h_D$
$H_G = W_{m1} - H$		$h_G = 0.5(C - G)$		$M_G = H_G h_G$
$H_T = H - H_D$		$h_T = 0.5(h_D + h_G)$		$M_T = H_T h_T$
				M_o
Seating				
$H_G = W$		$h_G = 0.5(C - G)$		M'_o
6 SHAPE CONSTANTS				
$K = A/B$		Y		
If bolt spacing exceeds $2a + t$, multiply M_o and M'_o in above equation by: $\sqrt{\frac{\text{Bolt spacing}}{2a + t}}$				
7 FLANGE THICKNESS REQUIRED				
t = greater of				
Operating		Seating		
$t = \sqrt{\frac{M_o Y}{S_{10} B}}$		$t = \sqrt{\frac{M'_o Y}{S_{1a} B}}$		

Figure 2-12. Dimensional data and forces for a ring flange.

Figure 2-13. Various attachments of ring flanges. (All other dimensions and loadings per Figure 2-11.)

8 NOTES	
If $g_o < 1.5t_n$ and $h < g_o$, design as integral. If $g_o > 1.5t_n$ and $h > g_o$, design as loose.	
If $g_o \leq 5/8$ in., $B/g_o \leq 300$, $P \leq 300$ psi and design temp. $< 700^\circ$, design as integral or loose.	
$c = \text{lesser of } t_n \text{ or } \begin{cases} \text{loose: } 2t_m \\ \text{integral: } 2g_o \end{cases}$ but not less than $1/4$ in.	

Adapted from Taylor Forge International, Inc., by permission.

TYPE 4: REVERSE FLANGE DESIGN

1 DESIGN CONDITIONS					
Design pressure, P		Allowable Stresses			
Design temperature		Flange		Bolting	
Flange material		Design temp., S_{t0}		Design temp., S_b	
Bolting material		Atm. temp., S_{t0}		Atm. temp., S_b	
Corrosion allowance					
2 GASKET AND FACING DETAILS					
Gasket		Facing			
3 TABLES 2-3 AND 2-4		4 LOAD AND BOLT CALCULATIONS			
N		$W_{m2} = b\pi Gy$		$A_m = \text{greater of } W_{m2}/S_a \text{ or } W_{m1}/S_b$	
b		$H_p = 2b\pi GmP$		A_b	
G		$H = G^2\pi P/4$		$W = 0.5(A_m + A_b)S_a$	
y		$W_{m1} = H_p + H$			
m					
5 MOMENT CALCULATIONS					
Load		x	Lever Arm		= Moment
Operating					
$H_D = \pi B^2 P/4$		$h_D = 0.5(C + g_1 - 2g_0 - B)$		$M_D = H_D h_D$	
$H_G = W_{m1} - H$		$h_G = 0.5(C - G)$		$M_G = H_G h_G$	
$H_T = H - H_D$		$h_T = 0.5(C - (B + G)/2)$		$M_T = H_T h_T$	
Add moments algebraically, then use the absolute value $ M_0 $ in all subsequent calculations.				$ M_0 $	
Seating					
$H_G = W$		$h_G = 0.5(C - G)$		M'_0	
6 K AND HUB FACTORS					
$K = A/B'$		h/h_0			
T		F			
Z		V			
Y		f			
U		$e = F/h_0$			
g_1/g_0		$d = \frac{U_R}{V} h_0 g_0^2$			
$h_0 = \sqrt{A g_0}$					
$Y_R = \alpha_R Y$		$U_R = \alpha_R U$			
$\alpha_R = \frac{1}{K^2} \left[1 + \frac{3(K+1)(1-\nu)}{\pi Y} \right]$					
$T_R = \frac{(Z+\nu)}{(Z-\nu)} \alpha_R T$					
7 STRESS FORMULA FACTORS					
t		$\delta = t^3/d$			
$\alpha = te + 1$		$\lambda = \gamma + \delta$			
$\beta = 4/3 te + 1$		$m_0 = M_0/B'$			
$\gamma = \alpha/T_R$		$m_G = M'_0/B'$			
8 STRESS CALCULATIONS					
Allowable Stress	Operating		Allowable Stress	Seating	
$1.5 S_{t0}$	Longitudinal hub, $S_H = fm_0/\lambda g_1^2$		$1.5 S_{t0}$	Longitudinal hub, $S_H = fm_0/\lambda g_1^2$	
S_{t0}	Radial flange, $S_R = \beta m_0/\lambda t^2$		S_{t0}	Radial flange, $S_R = \beta m_0/\lambda t^2$	
S_{t0}	Tangential flange, $S_T = m_0 Y_R/\lambda^2 - Z S_R$ ($0.67te + 1$)/ β		S_{t0}	Tangential flange, $S_T = m_0 Y_R/\lambda^2 - Z S_R$ ($0.67te + 1$)/ β	
S_{t0}	Greater of $0.5(S_H + S_R)$ or $0.5(S_H + S_T)$		S_{t0}	Greater of $0.5(S_H + S_R)$ or $0.5(S_H + S_T)$	
S_{t0}	Tangential flange $S_T(A_T B')$ $= \frac{m_0}{t^2}$ $\left[Y - \frac{2k^2 \left(1 + \frac{2}{3} te \right)}{(k^2 - 1)\lambda} \right]$		S_{t0}	Tangential flange $S_T(A_T B') = \frac{m_G}{t^2}$ $\left[Y - \frac{2k^2 \left(1 + \frac{2}{3} te \right)}{(k^2 - 1)\lambda} \right]$	

Figure 2-14. Dimensional data and forces for a reverse flange.

Adapted from Taylor Forge International, Inc., by permission.

TYPE 5: SLIP-ON FLANGE, FLAT FACE, FULL GASKET

1 DESIGN CONDITIONS					
Design pressure, P		Allowable Stresses			
Design temperature		Flange		Bolting	
Flange material		Design temp., S_{fo}		Design temp., S_b	
Bolting material		Atm. temp., S_{fa}		Atm. temp., S_a	
Corrosion allowance					
2 GASKET AND FACING DETAILS					
Gasket		Facing			
3 TABLES 2-3 AND 2-4		4 LOAD AND BOLT CALCULATIONS			
$G = C - 2h_g$		$W_{m2} = b\pi Gy + H'_{gy}$		$A_m = \text{greater of } W_{m2}/S_a \text{ or } W_{m1}/S_b$	
$b = (C - B)/4$		$H_p = 2b\pi GmP$		A_b	
y		$H'_p = (h_g/h'_g)H_p$		$W = 0.5(A_m + A_b)S_a$	
m		$H = G^2\pi P/4$		$H'_{gy} = (h_g/h'_g)b\pi Gy$	
		$W_{m1} = H + H_p + H'_p$			
5 MOMENT CALCULATIONS					
Load		x Lever Arm		= Moment	
Operating					
$H_D = \pi B^2 P/4$		$h_D = R + g_1$		$M_D = H_D h_D$	
$H_T = H - H_D$		$h_T = 0.5(R + g_1 + h_g)$		$M_T = H_T h_T$	
				M_o	
Lever Arms					
$h_g = \frac{(C - B)(2B + C)}{6(B + C)}$		$h'_g = \frac{(A - C)(2A + C)}{6(C + A)}$			
Reverse Moment					
$H_G = W - H$		$h'_G = \frac{h_g h'_g}{h_g + h'_g}$		$M_G = H_G h'_G$	
6 K AND HUB FACTORS					
$K = A/B$		h/h_o			
T		F_L			
Z		V_L			
Y		$e = \frac{F_L}{h_o}$			
U		$d = \frac{U}{V_L} h_o g_o^2$			
g_1/g_o					
$h_o = \sqrt{B g_o}$					
7 STRESS FORMULA FACTORS					
t		$\delta = t^3/d$			
$\alpha = te + 1$		$\lambda = \gamma + \delta$			
$\beta = 4/3 te + 1$		$m_o = M_o/B$			
$\gamma = \alpha/T$					
If bolt spacing exceeds $2a + t$, multiply m_o in above equation by: $\sqrt{\frac{\text{Bolt spacing}}{2a + t}}$					
8 STRESS CALCULATIONS					
Allowable Stress	Operating				
$1.5 S_{fo}$	Longitudinal hub, $S_H = m_o/\lambda g_1^2$				
S_{fo}	Radial flange, $S_R = \beta m_o/\lambda t^2$				
S_{fo}	Tangential flange, $S_T = m_o Y/t^2 - Z S_R$				
S_{fo}	Greater of $0.5(S_H + S_R)$ or $0.5(S_H + S_T)$				
S_{fo}	Radial stress at bolt circle $S_{RAD} = \frac{6M_G}{t^2(\pi C - \pi d_1)}$				

Figure 2-15. Dimensional data and forces for a slip-on flange, flat face, full gasket.

Adapted from Taylor Forge International, Inc., by permission.

Table 2-3
Gasket Materials and Contact Facings¹
Gasket Factors (m) for Operating Conditions and Minimum Design Seating Stress (y)

Gasket Material		Gasket Factor m	Min. Design Seating Stress y	Sketches and Notes	Use Facing Sketch Refer to Table 2-4	Use Column
Self-energizing types: O rings, metallic, elastomer or other gasket types considered as self-sealing		0	0			
Elastomers without fabric or a high percentage of asbestos fiber: Below 75A Shore Durometer 75A or higher Shore Durometer		0.50 1.00	0 200		(1a), (1b), (1c), (1d), (4), (5)	II
Asbestos with a suitable binder for the operating conditions	1/8 thick 1/16 thick 1/32 thick	2.00 2.75 3.50	1,600 3,700 6,500			
Elastomers with cotton fabric insertion		1.25	400			
Elastomers with asbestos fabric insertion, with or without wire reinforcement	3-ply 2-ply 1-ply	2.25 2.50 2.75	2,200 2,900 3,700	  		
Vegetable fiber		1.75	1,100			
Spiral-wound metal, asbestos filled	Carbon Stainless or Monel	2.50 3.00	10,000 10,000			
Corrugated metal, asbestos inserted or corrugated metal, jacketed asbestos filled	Soft aluminum Soft copper or brass Iron or soft steel Monel or 4%-6% chrome Stainless steels	2.50 2.75 3.00 3.25 3.50	2,900 3,700 4,500 5,500 6,500	 		
Corrugated metal	Soft aluminum Soft copper or brass Iron or soft steel Monel or 4%-6% chrome Stainless steels	2.75 3.00 3.25 3.50 3.75	3,700 4,500 5,500 6,500 7,600		(1a), (1b), (1c), (1d)	
Flat metal jacketed asbestos filled	Soft aluminum Soft copper or brass Iron or soft steel Monel or 4%-6% chrome Stainless steels	3.25 3.50 3.75 3.50 3.75 3.75	5,500 6,500 7,600 8,000 9,000 9,000	  	(1a), (1b), (1c), (1d), (2) ²	
Grooved metal	Soft aluminum Soft copper or brass Iron or soft steel Monel or 4%-6% chrome Stainless steels	3.25 3.50 3.75 3.75 4.25	5,500 6,500 7,600 9,000 10,100		(1a), (1b), (1c), (1d), (2), (3)	
Solid flat metal	Soft aluminum Soft copper or brass Iron or soft steel Monel or 4%-6% chrome Stainless steels	4.00 4.75 5.50 6.00 6.50	8,800 13,000 18,000 21,800 26,000		(1a), (1b), (1c), (1d), (2), (3), (4), (5)	I
Ring joint	Iron or soft steel Monel or 4%-6% chrome Stainless steels	5.50 6.00 6.50	18,000 21,800 26,000		(6)	

NOTES:

- This table gives a list of many commonly used gasket materials and contact facings with suggested design values of m and y that have generally proved satisfactory in actual service when using effective gasket seating width b given in Table 2-4. The design values and other details given in this table are suggested only and are not mandatory.
- The surface of a gasket having a lap should not be against the nubbin.

Reprinted by permission from ASME Code Section VIII Div. 1, Table 2-5.1.

Table 2-4
Effective Gasket Width

Facing Sketch (Exaggerated)		Basic Gasket Seating Width, b_o	
		Column I	Column II
(1a)		$\frac{N}{2}$	$\frac{N}{2}$
(1b) ¹			
(1c)		$\frac{w + T}{2} \left(\frac{w + N}{4} \max \right)$	$\frac{w - T}{2} \left(\frac{w - N}{4} \max \right)$
(1d)			
(2)		$\frac{w + N}{4}$	$\frac{w + 3N}{8}$
(3)		$\frac{N}{4}$	$\frac{3N}{8}$
(4) [*]		$\frac{3N}{8}$	$\frac{7N}{16}$
(5) [*]		$\frac{N}{4}$	$\frac{3N}{8}$
(6)		$\frac{w}{8}$	
Effective Gasket Seating Width, b			
$b = b_o$, when $b_o \leq \frac{1}{4}$ in.			
$b = \frac{\sqrt{b_o}}{2}$, when $b_o > \frac{1}{4}$ in.			
Location of Gasket Load Reaction			
For $b_o > \frac{1}{4}$ in.		For $b_o \leq \frac{1}{4}$ in.	<i>Note:</i> The gasket factors listed only apply to flanged joints in which the gasket is contained entirely within the inner edges of the bolt holes

^{*} Where serrations do not exceed $\frac{1}{64}$ -in. depth and $\frac{1}{32}$ -in. width spacing, sketches (1b) and (1d) shall be used.

Reprinted by permission from ASME Code Section VIII Div. 1, Table 2-5.2.

Table 2-5
Table of Coefficients

K	T	Z	Y	U	K	T	Z	Y	U	K	T	Z	Y	U	K	T	Z	Y	U
1.001	1.91	1000.50	1911.16	2100.18	1.046	1.90	22.05	42.75	46.99	1.091	1.88	11.52	22.22	24.41	1.136	1.86	7.88	15.26	16.77
1.002	1.91	500.50	956.16	1050.72	1.047	1.90	21.79	41.87	46.03	1.092	1.88	11.40	21.99	24.16	1.137	1.86	7.83	15.15	16.65
1.003	1.91	333.83	637.85	700.93	1.048	1.90	21.35	41.02	45.09	1.093	1.88	11.28	21.76	23.91	1.138	1.86	7.78	15.05	16.54
1.004	1.91	250.50	478.71	526.05	1.049	1.90	20.92	40.21	44.21	1.094	1.88	11.16	21.54	23.67	1.139	1.86	7.73	14.95	16.43
1.005	1.91	200.50	383.22	421.12	1.050	1.89	20.51	39.43	43.34	1.095	1.88	11.05	21.32	23.44	1.140	1.86	7.68	14.86	16.35
1.006	1.91	167.17	319.56	351.16	1.051	1.89	20.12	38.68	42.51	1.096	1.88	10.94	21.11	23.20	1.141	1.86	7.62	14.76	16.22
1.007	1.91	143.36	274.09	301.20	1.052	1.89	19.74	37.96	41.73	1.097	1.88	10.83	20.91	22.97	1.142	1.86	7.57	14.66	16.11
1.008	1.91	125.50	239.95	263.75	1.053	1.89	19.38	37.27	40.96	1.098	1.88	10.73	20.71	22.75	1.143	1.86	7.53	14.57	16.01
1.009	1.91	111.61	213.40	234.42	1.054	1.89	19.03	36.60	40.23	1.099	1.88	10.62	20.51	22.39	1.144	1.86	7.48	14.48	15.91
1.010	1.91	100.50	192.19	211.19	1.055	1.89	18.69	35.96	39.64	1.100	1.88	10.52	20.31	22.18	1.145	1.86	7.43	14.39	15.83
1.011	1.91	91.41	174.83	192.13	1.056	1.89	18.38	35.34	38.84	1.101	1.88	10.43	20.15	22.12	1.146	1.86	7.38	14.29	15.71
1.012	1.91	83.84	160.38	176.25	1.057	1.89	18.06	34.74	38.19	1.102	1.88	10.33	19.94	21.92	1.147	1.86	7.34	14.20	15.61
1.013	1.91	77.43	148.06	162.81	1.058	1.89	17.76	34.17	37.56	1.103	1.88	10.23	19.76	21.72	1.148	1.86	7.29	14.12	15.51
1.014	1.91	71.93	137.69	151.30	1.059	1.89	17.47	33.62	36.95	1.104	1.88	10.14	19.58	21.52	1.149	1.86	7.25	14.03	15.42
1.015	1.91	67.17	128.61	141.33	1.060	1.89	17.18	33.04	36.34	1.105	1.88	10.05	19.38	21.30	1.150	1.86	7.20	13.95	15.34
1.016	1.90	63.00	120.56	132.49	1.061	1.89	16.91	32.55	35.78	1.106	1.88	9.96	19.33	21.14	1.151	1.86	7.16	13.86	15.23
1.017	1.90	59.33	111.98	124.81	1.062	1.89	16.64	32.04	35.21	1.107	1.87	9.87	19.07	20.96	1.152	1.86	7.11	13.77	15.14
1.018	1.90	56.06	107.36	118.00	1.063	1.89	16.40	31.55	34.68	1.108	1.87	9.78	18.90	20.77	1.153	1.86	7.07	13.69	15.05
1.019	1.90	53.14	101.72	111.78	1.064	1.89	16.15	31.08	34.17	1.109	1.87	9.70	18.74	20.59	1.154	1.86	7.03	13.61	14.96
1.020	1.90	50.51	96.73	106.30	1.065	1.89	15.90	30.61	33.65	1.110	1.87	9.62	18.55	20.38	1.155	1.86	6.99	13.54	14.87
1.021	1.90	48.12	92.21	101.33	1.066	1.89	15.67	30.17	33.17	1.111	1.87	9.54	18.42	20.25	1.156	1.86	6.95	13.45	14.78
1.022	1.90	45.96	88.04	96.75	1.067	1.89	15.45	29.74	32.69	1.112	1.87	9.46	18.27	20.08	1.157	1.86	6.91	13.37	14.70
1.023	1.90	43.98	84.30	92.64	1.068	1.89	15.22	29.32	32.22	1.113	1.87	9.38	18.13	19.91	1.158	1.86	6.87	13.30	14.61
1.024	1.90	42.17	80.81	88.81	1.069	1.89	15.02	28.91	31.79	1.114	1.87	9.30	17.97	19.75	1.159	1.86	6.83	13.22	14.53
1.025	1.90	40.51	77.61	85.29	1.070	1.89	14.80	28.51	31.34	1.115	1.87	9.22	17.81	19.55	1.160	1.86	6.79	13.15	14.45
1.026	1.90	38.97	74.70	82.09	1.071	1.89	14.61	28.13	30.92	1.116	1.87	9.15	17.68	19.43	1.161	1.85	6.75	13.07	14.36
1.027	1.90	37.54	71.97	79.08	1.072	1.89	14.41	27.76	30.51	1.117	1.87	9.07	17.54	19.27	1.162	1.85	6.71	13.00	14.28
1.028	1.90	36.22	69.43	76.30	1.073	1.89	14.22	27.39	30.11	1.118	1.87	9.00	17.40	19.12	1.163	1.85	6.67	12.92	14.20
1.029	1.90	34.99	67.11	73.75	1.074	1.88	14.04	27.04	29.72	1.119	1.87	8.94	17.27	18.98	1.164	1.85	6.64	12.85	14.12
1.030	1.90	33.84	64.91	71.33	1.075	1.88	13.85	26.69	29.34	1.120	1.87	8.86	17.13	18.80	1.165	1.85	6.60	12.78	14.04
1.031	1.90	32.76	62.85	69.06	1.076	1.88	13.68	26.36	28.98	1.121	1.87	8.79	17.00	18.68	1.166	1.85	6.56	12.71	13.97
1.032	1.90	31.76	60.92	66.94	1.077	1.88	13.56	26.03	28.69	1.122	1.87	8.72	16.87	18.54	1.167	1.85	6.53	12.64	13.89
1.033	1.90	30.81	59.11	64.95	1.078	1.88	13.35	25.72	28.27	1.123	1.87	8.66	16.74	18.40	1.168	1.85	6.49	12.58	13.82
1.034	1.90	29.92	57.41	63.08	1.079	1.88	13.18	25.40	27.92	1.124	1.87	8.59	16.62	18.26	1.169	1.85	6.46	12.51	13.74
1.035	1.90	29.08	55.80	61.32	1.080	1.88	13.02	25.10	27.59	1.125	1.87	8.53	16.49	18.11	1.170	1.85	6.42	12.43	13.66
1.036	1.90	28.29	54.29	59.66	1.081	1.88	12.87	24.81	27.27	1.126	1.87	8.47	16.37	17.99	1.171	1.85	6.39	12.38	13.60
1.037	1.90	27.54	52.85	58.08	1.082	1.88	12.72	24.52	26.95	1.127	1.87	8.40	16.25	17.86	1.172	1.85	6.35	12.31	13.53
1.038	1.90	26.83	51.50	56.59	1.083	1.88	12.57	24.24	26.65	1.128	1.87	8.34	16.14	17.73	1.173	1.85	6.32	12.25	13.46
1.039	1.90	26.15	50.21	55.17	1.084	1.88	12.43	24.00	26.34	1.129	1.87	8.28	16.02	17.60	1.174	1.85	6.29	12.18	13.39
1.040	1.90	25.51	48.97	53.82	1.085	1.88	12.29	23.69	26.05	1.130	1.87	8.22	15.91	17.48	1.175	1.85	6.25	12.10	13.30
1.041	1.90	24.90	47.81	53.10	1.086	1.88	12.15	23.44	25.77	1.131	1.87	8.16	15.79	17.35	1.176	1.85	6.22	12.06	13.25
1.042	1.90	24.32	46.71	51.33	1.087	1.88	12.02	23.18	25.48	1.132	1.87	8.11	15.68	17.24	1.177	1.85	6.19	12.00	13.18
1.043	1.90	23.77	45.64	50.15	1.088	1.88	11.89	22.93	25.20	1.133	1.86	8.05	15.57	17.11	1.178	1.85	6.16	11.93	13.11
1.044	1.90	23.23	44.64	49.05	1.089	1.88	11.76	22.68	24.93	1.134	1.86	7.99	15.46	16.99	1.179	1.85	6.13	11.87	13.05
1.045	1.90	22.74	43.69	48.02	1.090	1.88	11.63	22.44	24.66	1.135	1.86	7.94	15.36	16.90	1.180	1.85	6.10	11.79	12.96